CBSE CURRICULUM 2024-25

Artificial Intelligence

Subject Code 417 — Class X

Complete Study Guide & Reference Book

Part A: Employability Skills • Part B: AI Concepts • Python • Data Science

Computer Vision • NLP • Evaluation • Practical Programs

Part C: Practical Exam Guide • Part D: Project Work

MCQs • Short & Long Answers • Viva Questions • Revision Notes

2025

Table of Contents

Artificial Intelligence (Subject Code 417) — Class X — Total: 100 Marks

PART A — EMPLOYABILITY SKILLS (50 HOURS | 10 MARKS)

Unit 1 **Communication Skills-II** — 10 hrs | 2 marks

Unit 2 **Self-Management Skills-II** — 10 hrs | 2 marks

Unit 3 **ICT Skills-II** — 10 hrs | 2 marks

Unit 4 **Entrepreneurial Skills-II** — 15 hrs | 2 marks

Unit 5 **Green Skills-II** — 5 hrs | 2 marks

PART B — SUBJECT SPECIFIC SKILLS (150 HOURS | 40 MARKS)

Unit 1 **Introduction to Artificial Intelligence (AI)** — 15 hrs theory | 7 marks

Unit 2 **AI Project Cycle** — 15 hrs theory | 9 marks

Unit 3 **Advance Python** — 30 hrs practical | assessed in practicals only

Unit 4 **Data Science** — 7 hrs theory + 8 hrs practical | 4 marks theory

Unit 5 **Computer Vision** — 12 hrs theory + 18 hrs practical | 4 marks theory

Unit 6 **Natural Language Processing** — 25 hrs theory + 5 hrs practical | 8 marks

Unit 7 **Evaluation** — 15 hrs | 8 marks

PART C — PRACTICAL WORK (35 MARKS)

- **Practical File (min 15 programs)** — 15 marks
- **Practical Examination (Python + Data Science + CV)** — 15 marks
- **Viva Voce** — 5 marks

PART D — PROJECT WORK / FIELD VISIT / STUDENT PORTFOLIO (15 MARKS)

- **Project / Field Visit / Portfolio** — 10 marks
- **Viva Voce** — 5 marks

Unit 1

Communication Skills-II

Master effective communication — verbal, non-verbal, and visual — for personal and professional success.

PART A — Employability Skills   10 Hours   2 Marks

1.1 Methods of Communication

Definition

Communication is the process of exchanging information, ideas, thoughts, or feelings between two or more people through a common system of symbols, signs, or behaviour.

Communication can be broadly classified into three types:

Type	Meaning	Examples
Verbal Communication	Communication using words — spoken or written	Face-to-face talk, phone call, letter, email, report
Non-Verbal Communication	Communication without words — body language, gestures, expressions	Smile, handshake, eye contact, posture, tone of voice
Visual Communication	Communication through visual elements	Charts, graphs, maps, posters, traffic signals, logos

Verbal Communication

Verbal communication uses words to share information. It is the most common method of communication and can be further divided into two types:

- **Oral Communication:** Speaking and listening. Examples: conversations, speeches, interviews, phone calls, video calls, group discussions.
- **Written Communication:** Using written words. Examples: letters, emails, reports, text messages, notices, articles, blogs.

Aspect	Oral Communication	Written Communication
Speed	Fast, immediate	Slower, takes time
Feedback	Instant feedback possible	Delayed feedback
Record	No permanent record (unless recorded)	Permanent record available
Formality	Can be informal	Usually formal
Best For	Quick discussions, emotional messages	Official matters, detailed information

Non-Verbal Communication

Research shows over 55% of communication is non-verbal. It includes:

- **Facial Expressions:** Smiling shows friendliness; frowning shows displeasure. The face can express happiness, sadness, anger, surprise, fear, and disgust.
- **Gestures:** Waving, pointing, thumbs up. Different cultures may interpret gestures differently.
- **Body Language/Posture:** Standing straight shows confidence; slouching may show disinterest. Leaning forward shows interest.
- **Eye Contact:** Maintaining eye contact shows attentiveness and honesty. Avoiding eye contact may signal nervousness or dishonesty.
- **Tone of Voice (Paralanguage):** Pitch, volume, speed, and pauses. The same words said in different tones can convey different meanings.

- **Personal Space (Proxemics):** Physical distance maintained during communication. Intimate (0-18 inches), Personal (18 inches-4 feet), Social (4-12 feet), Public (12+ feet).
- **Touch (Haptics):** Handshakes, pats on the back, hugs convey emotions and relationships.

Real-Life Example

Imagine a teacher is explaining a lesson. A student sits with arms crossed, looks away, and yawns. Even though the student has not spoken a word, the non-verbal cues tell the teacher that the student is bored or uninterested.

Visual Communication

Visual communication uses images, symbols, and graphics to convey messages. It is often faster and more effective than words alone. Examples include:

- Charts, graphs, and diagrams in reports and textbooks
- Traffic signs and road signals
- Logos and brand symbols (e.g., Nike swoosh)
- Maps and infographics
- Emojis in digital communication
- Posters, banners, and advertisements

Exam Tip

Questions often ask you to differentiate between verbal and non-verbal communication. Remember: verbal = words (spoken or written), non-verbal = everything else (gestures, expressions, tone). Visual communication is a separate category that uses images, charts, and symbols.

1.2 Writing Skills — Sentences and Parts of Speech

Types of Sentences

Type	Purpose	Punctuation	Example
Declarative	States a fact or information	Full stop (.)	AI is transforming industries.
Interrogative	Asks a question	Question mark (?)	What is machine learning?
Imperative	Gives a command or request	Full stop (.) or exclamation (!)	Please submit your project by Friday.
Exclamatory	Expresses strong emotion	Exclamation mark (!)	What an amazing invention AI is!

Parts of Speech

There are eight parts of speech in English. Every word in a sentence belongs to one of these categories:

Part of Speech	Function	Example
Noun	Names a person, place, thing, or idea	Ravi, Delhi, computer, happiness
Pronoun	Replaces a noun	he, she, it, they, we
Verb	Shows action or state of being	run, write, is, think
Adjective	Describes a noun	smart, tall, beautiful, three
Adverb	Describes a verb, adjective, or adverb	quickly, very, always, here
Preposition	Shows relationship between words	in, on, at, between, under
Conjunction	Joins words, phrases, or clauses	and, but, or, because, although
Interjection	Expresses emotion	Wow! Ouch! Hurray!

Sentence Breakdown

"*The clever student quickly solved the difficult problem.*"
The = Article, clever = Adjective, student = Noun, quickly = Adverb, solved = Verb, the = Article, difficult = Adjective, problem = Noun.

Articles

- **A** — before consonant sounds: *a book, a university* (indefinite article)
- **An** — before vowel sounds: *an apple, an hour* (indefinite article)
- **The** — definite article for specific nouns: *The Taj Mahal, the sun*

> **Memory Trick — A vs. An**
> It is the **sound**, not the **letter**, that decides. "A university" (consonant sound "yu") but "An umbrella" (vowel sound "uh"). "An hour" (silent 'h', vowel sound) but "A hotel" (pronounced 'h', consonant sound).

Subject-Verb Agreement

The verb in a sentence must agree with its subject in number (singular or plural):

- Singular subject takes singular verb: "The boy **runs** fast."
- Plural subject takes plural verb: "The boys **run** fast."
- "I" always takes a plural verb: "I **am** a student." (not "I is")

1.3 Active and Passive Voice

> **Definition**
> **Active Voice:** The subject performs the action. *"The programmer wrote the code."*
> **Passive Voice:** The subject receives the action. *"The code was written by the programmer."*

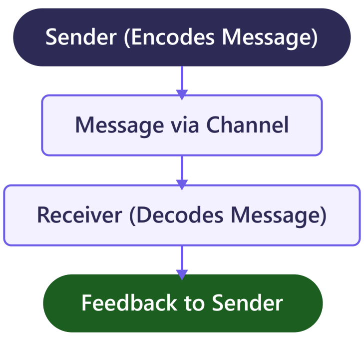
In active voice, the doer of the action comes first. In passive voice, the object of the action comes first. To convert from active to passive: (1) Object becomes subject, (2) Verb changes to "be + past participle", (3) Subject becomes "by + agent" (can be dropped).

Tense	Active Voice	Passive Voice
Simple Present	She teaches AI.	AI is taught by her.
Simple Past	He wrote a program.	A program was written by him.
Present Continuous	They are building a robot.	A robot is being built by them.
Past Continuous	She was reading a book.	A book was being read by her.
Present Perfect	He has completed the task.	The task has been completed by him.
Future Simple	They will launch the app.	The app will be launched by them.

> **When to Use Which?**
> Use **Active Voice** when you want to emphasise who did the action. Use **Passive Voice** when the action is more important than the doer, or when the doer is unknown. Example: "The window was broken." (doer unknown).

1.4 Communication Cycle & Feedback

The communication cycle is the complete process through which a message travels from sender to receiver and back. It is a continuous process.



Elements of the Communication Cycle

Element	Description	Example
Sender	The person who initiates the message	A teacher explaining a topic
Encoding	Converting thoughts into words, symbols, or gestures	Teacher organises thoughts into sentences
Message	The actual content being communicated	The lesson content
Channel	The medium through which the message is sent	Speech (oral), email (written), video call
Receiver	The person who receives the message	Students in the classroom
Decoding	Interpreting and understanding the message	Students understanding the lesson
Feedback	The receiver's response back to the sender	Students asking questions or nodding

Types of Feedback

- **Positive Feedback:** Encourages the sender. Example: "Great presentation!"
- **Negative Feedback:** Points out problems. Example: "Your report has many errors."
- **Constructive Feedback:** Suggests improvements. Example: "Your report is good, but adding graphs would make it better."

Why is Feedback Important?

Feedback confirms the message was understood correctly, helps improve future communication, builds trust between sender and receiver, and completes the communication cycle. Without feedback, communication is one-way and incomplete.

1.5 Barriers to Effective Communication

Barriers are obstacles that prevent the message from being understood correctly. They can occur at any point in the communication cycle.

Barrier	Description	Examples
Physical Barriers	Environmental obstacles that interfere with communication	Noise, poor lighting, distance, faulty equipment, bad internet connection
Linguistic Barriers	Language differences, jargon, unclear words	Technical words with non-technical audience, different languages, slang
Psychological Barriers	Mental or emotional factors that block communication	Stress, anger, fear, prejudice, lack of attention, poor memory
Cultural Barriers	Differences in customs, values, and beliefs	Gestures meaning different things in different cultures, dietary differences, dress codes
Organisational Barriers	Problems in the structure of an organisation	Too many levels of management, strict rules, lack of trust

How to Overcome Communication Barriers

- Use simple and clear language that the audience can understand
- Be an active listener — pay full attention, do not interrupt
- Give and seek feedback to ensure the message is understood
- Choose the right channel for the message (serious matters should be in writing)
- Be aware of cultural differences and show respect
- Control emotions — do not communicate when angry or stressed
- Reduce physical barriers — find a quiet place, use good equipment
- Use visual aids (charts, diagrams) to support the message

Real-Life Example

A company has employees from India, Japan, and the USA in a video meeting. The Indian manager uses the phrase "do the needful" which is common in India but confusing for others. This is a linguistic and cultural barrier. The solution is to use universally understood phrases like "please complete the required task."

Chapter Summary

- Communication is the exchange of information between people through verbal, non-verbal, and visual methods.
- Verbal communication uses words (oral and written). Non-verbal uses body language, gestures, and tone.
- There are four types of sentences: Declarative, Interrogative, Imperative, and Exclamatory.
- Eight parts of speech: Noun, Pronoun, Verb, Adjective, Adverb, Preposition, Conjunction, Interjection.
- Active voice: subject does the action. Passive voice: subject receives the action.
- The communication cycle includes Sender, Encoding, Message, Channel, Receiver, Decoding, and Feedback.
- Barriers to communication: Physical, Linguistic, Psychological, Cultural, and Organisational.
- Feedback is essential to complete the communication cycle and ensure understanding.

Communication | Verbal | Non-Verbal | Visual | Active Voice | Passive Voice | Feedback | Barriers | Communication Cycle | Parts of Speech | Proxemics | Paralanguage | Encoding | Decoding

MCQs

1 Which is an example of non-verbal communication?

a) Writing a letter

b) Sending an email

c) Maintaining eye contact

d) Making a phone call

Answer: c) Eye contact is non-verbal as it does not use words.

2 "Please close the door." is which type of sentence?

a) Declarative

b) Interrogative

c) Exclamatory

d) Imperative

Answer: d) It gives a polite command or request.

3 In the communication cycle, the receiver's response is called:

a) Encoding

b) Channel

c) Feedback

d) Decoding

Answer: c) Feedback is the response sent back by the receiver.

4 Which of the following is a physical barrier to communication?

a) Prejudice

b) Anger

c) Loud noise in the room

d) Use of jargon

Answer: c) Loud noise is a physical (environmental) barrier. Prejudice and anger are psychological barriers. Jargon is a linguistic barrier.

5 "The code was written by the programmer." This sentence is in:

a) Active voice

b) Passive voice

c) Imperative voice

d) Exclamatory voice

Answer: b) The subject "code" is receiving the action. The doer "programmer" comes after "by", which is the passive voice structure.

6 The study of physical distance in communication is called:

a) Haptics

b) Proxemics

c) Paralanguage

d) Kinesics

Answer: b) Proxemics is the study of personal space and physical distance in communication.

7 Which article should be used: "____ honest person"?

a) A

b) An

c) The

d) No article

Answer: b) "Honest" starts with a silent 'h', so the sound is a vowel sound "o". Therefore we use "An honest person."

Short Answer Questions

Q1. What is the difference between verbal and non-verbal communication?

Answer: Verbal communication uses words (spoken or written) like conversations, letters, and emails. Non-verbal communication uses body language, facial expressions, gestures, eye contact, and tone of voice without using words. For example, saying "I am happy" is verbal communication while smiling is non-verbal communication. Research shows that over 55% of our communication is non-verbal.

Q2. What is feedback? Why is it important?

Answer: Feedback is the receiver's response to the sender after receiving the message. It is important because: (i) It confirms the message was understood correctly, (ii) It helps improve future communication, (iii) It builds trust between the sender and receiver, (iv) It completes the communication cycle, making it two-way, and (v) It allows the sender to correct any misunderstandings.

Q3. Convert the following sentences from active to passive voice: (a) She writes a letter. (b) They are building a house.

Answer: (a) Active: She writes a letter. Passive: A letter is written by her. (b) Active: They are building a house. Passive: A house is being built by them. In passive voice, the object of the active sentence becomes the subject, and the verb changes to "be + past participle."

Q4. List any four types of non-verbal communication with examples.

Answer: (i) **Facial Expressions:** A smile shows friendliness; a frown shows displeasure. (ii) **Gestures:** Waving hello, thumbs up for approval. (iii) **Eye Contact:** Looking at someone while speaking shows attentiveness and confidence. (iv) **Body Language/Posture:** Standing straight shows confidence; slouching may indicate boredom or lack of interest.

Q5. What are the different types of sentences? Give one example of each.

Answer: There are four types: (i) **Declarative** — states a fact: "The Earth revolves around the Sun." (ii) **Interrogative** — asks a question: "What is your name?" (iii) **Imperative** — gives a command or request: "Please sit down." (iv) **Exclamatory** — expresses strong emotion: "What a beautiful painting!"

Long Answer Questions

Q1. Explain the communication cycle with its key elements.

Answer: The communication cycle is the complete process through which a message is sent from sender to receiver and a response is received. It has the following key elements:

(1) **Sender:** The person who initiates and sends the message. The sender has an idea or thought to share.
(2) **Encoding:** The sender converts thoughts into words, symbols, or gestures that can be understood. For example, a teacher organises thoughts into sentences.
(3) **Message:** The actual content or information being communicated.
(4) **Channel:** The medium or method used to send the message — speech, email, phone, letter, or video call.
(5) **Receiver:** The person who receives the message.
(6) **Decoding:** The receiver interprets and tries to understand the message. Understanding depends on the receiver's knowledge, language skills, and attention.
(7) **Feedback:** The receiver sends a response back to the sender. This could be a reply, a question, a nod, or even silence.

The cycle is continuous — once feedback is given, the roles of sender and receiver may switch. Effective communication depends on all elements working properly. If any element fails (for example, a noisy channel or wrong decoding), the message may be misunderstood. Barriers such as noise, language differences, or emotional state can disrupt the cycle at any point.

Q2. What are barriers to effective communication? Explain each type with examples and suggest ways to overcome them.

Answer: Barriers to communication are obstacles that prevent the message from being clearly understood. The main types are:

(1) **Physical Barriers:** These are environmental factors such as noise, poor lighting, distance between people, faulty phone lines, or bad internet connections. Example: A student cannot hear the teacher clearly because of construction noise outside the classroom.
(2) **Linguistic Barriers:** These occur when people speak different languages, use technical jargon, or use ambiguous words. Example: A doctor using medical terms with a patient who does not understand them.
(3) **Psychological Barriers:** Mental and emotional states such as stress, anger, fear, prejudice, or lack of attention. Example: A student who is anxious about exams cannot concentrate on what the teacher is saying.
(4) **Cultural Barriers:** Differences in customs, traditions, beliefs, and social norms. Example: A thumbs-up gesture is positive in India but can be offensive in some Middle Eastern countries.
(5) **Organisational Barriers:** Issues within an organisation like too many management levels, rigid rules, or lack of trust. Example: Important messages get distorted as they pass through many levels of management.

Ways to Overcome: Use simple and clear language; be an active listener; give and seek feedback; choose the right communication channel; be culturally sensitive; control emotions before communicating; use visual aids to support the message; and ensure the physical environment is suitable for communication.

Unit 2

Self-Management Skills-II

Manage stress, time, emotions, and goals for personal and professional success.

PART A — Employability Skills   10 Hours   2 Marks

2.1 Meaning of Self-Management

Definition

Self-management is the ability to regulate and control one's own emotions, behaviour, thoughts, and actions in a productive way. It means taking responsibility for your own conduct and well-being.

Self-management is a vital skill for success in studies, career, and personal life. A person with good self-management skills can handle challenges calmly, stay focused, and work towards goals without needing constant supervision.

Key Components of Self-Management:

- **Self-Awareness:** Knowing your strengths, weaknesses, emotions, and their impact
- **Self-Motivation:** Internal drive to achieve goals without external pressure
- **Self-Regulation:** Controlling impulses and managing emotions appropriately
- **Time Management:** Planning and using time effectively
- **Stress Management:** Handling pressure without losing balance
- **Anger Management:** Controlling anger and responding calmly
- **Goal Setting:** Setting clear and achievable targets
- **Emotional Intelligence:** Understanding and managing your own and others' emotions

2.2 Stress Management

Definition

Stress is the body's response to any demand or challenge. It is a feeling of emotional or physical tension that arises when we face situations that test our ability to cope.

Types of Stress

Type	Description	Effect	Example
Eustress (Positive Stress)	Short-term stress that motivates and helps performance	Increases focus, energy, and performance	Excitement before a sports competition, nervousness before a stage performance
Distress (Negative Stress)	Prolonged or overwhelming stress that causes anxiety and harms health	Decreases performance, causes health problems	Constant exam worry, long-term bullying, family problems

Causes of Stress (Stressors)

- **Academic Pressure:** Exams, homework overload, competition among peers
- **Social Pressure:** Peer pressure, bullying, social media comparison
- **Family Issues:** Arguments at home, expectations from parents, financial problems
- **Health Problems:** Illness, lack of sleep, poor diet
- **Change:** Moving to a new school, loss of a friend, changes in routine
- **Technology:** Excessive screen time, cyberbullying, information overload

Signs of Stress

Physical Signs	Emotional Signs	Behavioural Signs

Headaches, body pain	Anxiety, irritability	Changes in eating habits
Fatigue, low energy	Sadness, mood swings	Difficulty sleeping
Stomach problems	Feeling overwhelmed	Withdrawing from friends
Rapid heartbeat	Lack of concentration	Procrastination

Stress Management Techniques (Detailed)

Technique	How It Helps	Practical Tip
Physical Exercise & Yoga	Releases endorphins (happy hormones), reduces tension	Walk, jog, or do yoga for 30 minutes daily
Deep Breathing	Activates the body's relaxation response, calms the mind	Breathe in for 4 counts, hold for 4, breathe out for 4 (Box Breathing)
Meditation	Reduces anxiety, improves focus and emotional control	Sit quietly for 10 minutes, focus on your breath
Time Management	Reduces the feeling of being overwhelmed	Use a planner, make to-do lists, prioritise tasks
Positive Thinking	Changes your perspective on challenges	Replace "I can't do this" with "I will try my best"
Talking to Someone	Sharing reduces the burden, provides support	Talk to a parent, teacher, friend, or counsellor
Adequate Sleep	Restores energy, improves mood and concentration	Sleep 7-8 hours every night, avoid screens before bed
Hobbies & Recreation	Provides a break from stress, brings joy	Read, paint, play music, garden, or play a sport
Healthy Diet	Proper nutrition supports brain and body function	Eat fruits, vegetables, whole grains; avoid junk food
Avoiding Procrastination	Starting tasks early reduces last-minute panic	Break large tasks into smaller steps, start with the easiest

Real-Life Example

Riya has her board exams in two weeks and feels extremely stressed. She starts waking up early, makes a study timetable, studies for 2-hour blocks with 15-minute breaks, goes for an evening walk, talks to her mother about her fears, and sleeps by 10 PM. Within a few days, she feels more confident and her stress reduces. This is effective stress management in action.

Remember the "4 A's" of Stress Management

Avoid unnecessary stress (say no to extra tasks), **Alter** the situation (change what you can), **Adapt** to the stressor (change your expectations), **Accept** things you cannot change.

2.3 Anger Management

Definition

Anger management is the process of learning to recognise the signs that you are becoming angry and taking action to deal with the situation in a positive and healthy way. Anger itself is a normal emotion, but uncontrolled anger can damage relationships and health.

Anger Management Techniques

- **Count to 10:** Before reacting, slowly count to 10. This gives your brain time to think logically.
- **Walk Away:** Remove yourself from the situation temporarily until you calm down.
- **Deep Breathing:** Take slow, deep breaths to reduce the physical response to anger.
- **Express Calmly:** Once calm, express your feelings using "I" statements: "I feel upset when..." instead of "You always..."
- **Physical Activity:** Channel the energy into exercise — go for a run, do push-ups.
- **Think Before Speaking:** Pause and think about what you want to say before blurting it out.
- **Identify Triggers:** Know what situations or people make you angry and prepare strategies in advance.

- **Seek Help:** If anger is frequent and uncontrollable, talk to a counsellor or trusted adult.

2.4 Self-Awareness & SWOT Analysis

Definition

Self-awareness is the ability to understand your own emotions, strengths, weaknesses, values, and their impact on others. It is the foundation of self-management.

SWOT Analysis (Detailed)

What is SWOT?

SWOT Analysis is a tool used to identify and analyse your **S**trengths, **W**eaknesses, **O**pportunities, and **T**hreats. It helps you understand where you stand and plan for improvement.

S — Strengths (Internal, Positive)	W — Weaknesses (Internal, Negative)
Things you are good at, your positive qualities, skills, and talents.	Areas where you need improvement, bad habits, or skills you lack.
Example: Good at mathematics, strong communication skills, hard-working	*Example: Procrastination, shy in public speaking, poor time management*
O — Opportunities (External, Positive)	**T — Threats (External, Negative)**
External chances that you can take advantage of for growth.	External challenges that could harm your progress.
Example: Scholarship exams, online courses, AI coding workshops	*Example: Social media distractions, peer pressure, competition*

Remember SWOT

Strengths and **W**eaknesses are **Internal** (within you). **O**pportunities and **T**hreats are **External** (outside you). Think: "SW = Inside Me, OT = Outside World"

2.5 Self-Motivation

Types of Motivation

Type	Source	Example	Lasting?
Intrinsic Motivation	Internal — personal interest, curiosity	Reading because you love reading; learning coding because it is fun	Long-lasting
Extrinsic Motivation	External — rewards, praise, grades	Studying hard to win a prize; cleaning room to get pocket money	Short-term

2.6 Time Management

Eisenhower Matrix (Priority Matrix)

	Urgent	Not Urgent
Important	**DO IT NOW** Exam tomorrow, medical emergency	**SCHEDULE IT** Regular revision, exercise, career planning
Not Important	**DELEGATE / DO QUICKLY** Routine messages, minor interruptions	**ELIMINATE / LIMIT** Excessive social media, random videos

Pomodoro Technique

Study for 25 minutes, take a 5-minute break, repeat. After 4 rounds, take a longer break (15-30 minutes).

2.7 Goal Setting — SMART Goals

Letter	Stands For	Meaning	Good Example

S	Specific	Clear and well-defined	"I will study Chapter 3 of AI"
M	Measurable	Track progress	"I will solve 20 MCQs daily"
A	Achievable	Realistic, not impossible	"I will improve my score by 10%"
R	Relevant	Aligns with larger objectives	"AI is relevant for my tech career"
T	Time-bound	Has a deadline	"Complete by 30th November"

2.8 Emotional Intelligence (EQ)

Definition

Emotional Intelligence (EQ) is the ability to recognise, understand, manage, and effectively use your own emotions and the emotions of others.

Five Components of Emotional Intelligence

Component	Meaning	Example
Self-Awareness	Knowing your own emotions and their effects	Realising you are angry before you react
Self-Regulation	Controlling your impulses and managing emotions	Taking a deep breath instead of shouting
Motivation	Internal drive to improve and achieve goals	Continuing to study despite failing a test
Empathy	Understanding and sharing others' feelings	Comforting a friend who is sad
Social Skills	Managing relationships and building networks	Working well in a team, resolving conflicts

Did You Know?

EQ is often a better predictor of success than IQ. Studies show that 90% of top performers at work have high emotional intelligence.

Chapter Summary & Keywords

Self-Management · Stress · Eustress · Distress · Anger Management · SWOT Analysis · SMART Goals · Emotional Intelligence · Time Management · Eisenhower Matrix · Pomodoro Technique · Intrinsic Motivation · Extrinsic Motivation · Empathy

MCQs

1 Excitement before a sports competition is an example of:

a) Distress

b) Eustress

c) Anxiety

d) Depression

Answer: b) Eustress is short-term positive stress that motivates.

2 "M" in SMART goals stands for:

a) Manageable

b) Meaningful

c) Measurable

d) Motivating

Answer: c) Measurable — you can track progress and know when achieved.

3 Understanding others' feelings is which component of EQ?

a) Self-awareness

b) Self-regulation

c) Empathy

d) Motivation

Answer: c) Empathy is the ability to understand and share others' feelings.

4 In SWOT Analysis, "Opportunities" refers to:

a) Your internal strengths

b) Your internal weaknesses

c) External favourable situations

d) External challenges

Answer: c) Opportunities are external factors you can use to your advantage.

5 The Pomodoro Technique involves studying for:

a) 10 minutes with 2-minute breaks

b) 25 minutes with 5-minute breaks

c) 60 minutes with 30-minute breaks

d) 45 minutes with 15-minute breaks

Answer: b) 25 minutes focused work + 5-minute break. After 4 rounds, longer break.

Short Answer Questions

Q1. Differentiate between Eustress and Distress.

Answer: Eustress is positive, short-term stress that motivates us and improves performance (e.g., excitement before a competition). **Distress** is negative, long-term stress that causes anxiety and harms health (e.g., constant exam worry). Eustress helps us grow, while distress needs to be managed.

Q2. What is SWOT Analysis? Explain with an example.

Answer: SWOT Analysis is a self-evaluation tool: **S**trengths (internal positive), **W**eaknesses (internal negative), **O**pportunities (external positive), **T**hreats (external negative). Example: A student's Strength = good at Maths, Weakness = poor English, Opportunity = free online courses, Threat = social media addiction.

Unit 3

ICT Skills-II

Use Information and Communication Technology tools effectively — OS, office apps, internet, networking, and cyber safety.

PART A — Employability Skills 10 Hours 2 Marks

3.1 Operating System (OS)

Definition

An Operating System (OS) is system software that manages computer hardware, software resources, and provides common services for application programs. It acts as an interface between the user and the hardware.

Functions: Process Management, Memory Management, File Management, Device Management, Security, User Interface (GUI/CLI).

Type	Used In	Examples
Desktop OS	Personal computers, laptops	Windows 10/11, macOS, Ubuntu (Linux)
Mobile OS	Smartphones, tablets	Android, iOS
Server OS	Servers handling multiple users	Windows Server, Linux
Embedded OS	Specific function devices	Smart TVs, ATMs, car systems

3.2 Office Applications

Word Processing (MS Word, Google Docs): Formatting text (bold/italic/underline), paragraphs, tables, headers/footers, spell check, find & replace, mail merge.

Spreadsheets (MS Excel, Google Sheets): Key formulas:

Formula	Purpose	Example
`=SUM(A1:A10)`	Adds values in a range	Sum of marks
`=AVERAGE(A1:A10)`	Calculates average	Average marks
`=MAX(A1:A10)`	Finds largest value	Highest marks
`=MIN(A1:A10)`	Finds smallest value	Lowest marks
`=COUNT(A1:A10)`	Counts cells with numbers	Number of entries
`=IF(A1>50,"Pass","Fail")`	Conditional result	Pass/Fail checker

Presentations (PowerPoint, Google Slides): Use large fonts (24pt+), bullet points, limit animations, follow 6x6 rule (max 6 lines, 6 words per line).

3.3 Email & Internet

Email parts: To, CC (Carbon Copy), BCC (Blind Carbon Copy), Subject, Body, Attachment, Signature.

Email etiquette: Clear subject line, proper greeting, concise professional message, proper closing, check attachments, don't use ALL CAPS.

Networking Basics

Type	Full Form	Range	Example
PAN	Personal Area Network	Few metres	Bluetooth connection
LAN	Local Area Network	Within a building	School computer lab

MAN	Metropolitan Area Network	Across a city	Cable TV network
WAN	Wide Area Network	Countries/continents	The Internet

3.4 Cyber Safety & Digital Footprint

Threat	Description	Prevention
Virus	Malicious software that attaches to files and spreads	Install antivirus, don't open unknown attachments
Phishing	Fake emails/websites to steal personal info	Check URL carefully, don't click suspicious links
Ransomware	Locks files and demands payment	Regular backups, never pay ransom
Trojan Horse	Malware disguised as useful software	Download only from trusted sources
Spyware	Secretly monitors your activities	Use anti-spyware, check app permissions
Cyberbullying	Using digital platforms to harass someone	Block, report, tell a trusted adult

Digital Footprint: Trail of data you leave online. **Active** (posts, emails) and **Passive** (browsing history, cookies). Think before you post — the internet has a long memory.

Internet Safety: Strong passwords (8+ chars, mixed), 2FA (Two-Factor Authentication), antivirus, secure websites (https://), log out on shared computers, avoid public Wi-Fi for sensitive tasks.

Keywords & MCQs

Operating System | GUI | CLI | Spreadsheet | Email | LAN | WAN | Wi-Fi | Cyber Safety | Phishing | Ransomware | Digital Footprint | 2FA

1 What is phishing?

a) A computer virus

b) Fake emails/websites to steal personal info

c) Encrypting data

d) A firewall

Answer: b) Phishing tricks users into revealing passwords, bank details, etc.

2 BCC in email stands for:

a) Basic Carbon Copy

b) Blind Carbon Copy

c) Bulk Carbon Copy

d) Best Carbon Copy

Answer: b) BCC recipients receive the email but their addresses are hidden from others.

3 Which formula finds the average in a spreadsheet?

a) =AVG(A1:A10)

b) =AVERAGE(A1:A10)

c) =MEAN(A1:A10)

d) =AVERAGE(A1-A10)

Answer: b) =AVERAGE(A1:A10) is the correct formula.

4 LAN stands for:

a) Large Area Network

b) Local Area Network

c) Long Area Network

d) Linked Area Network

Answer: b) Local Area Network — connects computers within a small area like a building.

5 Malware that locks files and demands money is called:

a) Virus

b) Trojan Horse

c) Ransomware

d) Spyware

Answer: c) Ransomware encrypts files and demands payment to unlock them.

Unit 4

Entrepreneurial Skills-II

Understand entrepreneurship, types, business planning, and government support schemes.

PART A — Employability Skills   15 Hours   2 Marks

4.1 Entrepreneurship

Definition

Entrepreneurship is the process of designing, launching, and running a new business, taking financial risks in hope of profit. An **entrepreneur** is the person who sets up this business.

Qualities: Creativity & Innovation, Risk-Taking, Self-Confidence, Hard Work & Persistence, Decision-Making, Leadership, Communication Skills, Time Management, Adaptability, Vision.

Types of Entrepreneurship

Type	Description	Example
Small Business	Local shops, restaurants, freelance	Neighbourhood grocery store
Scalable Start-up	Businesses that can grow rapidly	Flipkart, Zomato, Ola
Social	Solving social problems, not just profit	Sulabh International
Green	Environmental sustainability focus	Tesla, biodegradable packaging
Digital	Operates through digital platforms	YouTube creators, app developers

Entrepreneur vs Manager

Aspect	Entrepreneur	Manager
Role	Creates and owns the business	Manages existing business for owner
Risk	Takes personal financial risks	No personal financial risk (salary)
Reward	Earns profit (unlimited or zero)	Earns fixed salary
Innovation	Creates new products/ideas	Implements existing plans

4.2 Business Plan & Government Support

Business plan components: (1) Executive Summary (2) Business Description (3) Market Analysis (4) Organisation & Management (5) Products/Services (6) Marketing Strategy (7) Financial Plan (8) Operational Plan.

Myths vs Reality

Myth	Reality
Entrepreneurs are born, not made	Can be learned through education and experience
You need a lot of money to start	Many succeeded with minimal capital; ideas matter more
You need a completely new idea	Most businesses improve existing products/services
Failure means the end	Failure is a learning opportunity

Government Schemes

Scheme	Purpose	Key Feature
Start-up India (2016)	Build ecosystem for start-ups	Tax exemptions, easy registration, funding support

MUDRA Yojana (2015)	Loans for small/micro businesses	Shishu (up to Rs 50K), Kishor (up to Rs 5L), Tarun (up to Rs 10L)
Skill India (2015)	Skill training for youth	PMKVY, free training in 250+ job roles
Atal Innovation Mission (2016)	Promote innovation in schools	Atal Tinkering Labs with 3D printers, robotics

Keywords & MCQs

Entrepreneurship · Business Plan · Innovation · Risk-Taking · Start-up India · MUDRA · Social Entrepreneurship · Green Entrepreneurship

1 An entrepreneur is a person who:

a) Works in government

b) Sets up a business taking financial risks for profit

c) Only invests in stocks

d) Manages someone else's business

Answer: b) An entrepreneur creates, owns, and manages a business venture.

2 MUDRA "Shishu" category provides loans up to:

a) Rs. 10,000

b) Rs. 50,000

c) Rs. 5 lakh

d) Rs. 10 lakh

Answer: b) Shishu = up to Rs. 50,000. Kishor = up to Rs. 5 lakh. Tarun = up to Rs. 10 lakh.

3 Atal Tinkering Labs are set up under:

a) Start-up India

b) MUDRA

c) Atal Innovation Mission

d) Digital India

Answer: c) ATLs are part of Atal Innovation Mission (AIM).

Unit 5

Green Skills-II

Sustainable development, green economy, climate change, and environmental responsibility.

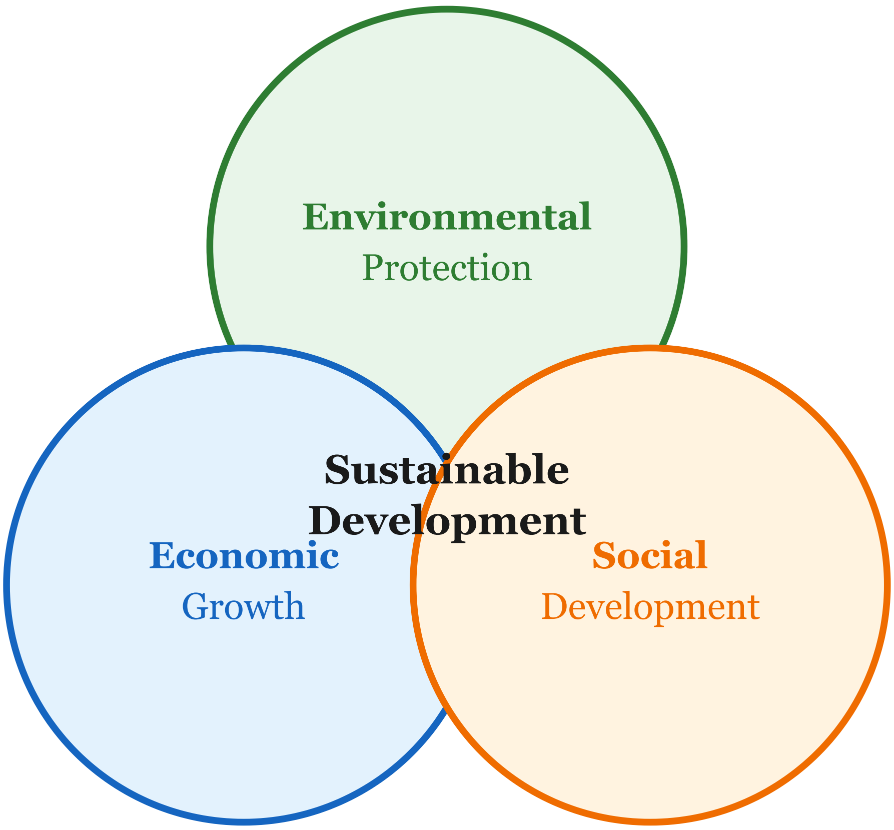
PART A — Employability Skills   5 Hours   2 Marks

5.1 Sustainable Development

Definition

Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs (Brundtland Commission, 1987).

Three Pillars: Economic Growth, Social Development, Environmental Protection.



Three Pillars of Sustainable Development

5.2 UN SDGs & Green Economy

The United Nations set **17 Sustainable Development Goals** in 2015, to be achieved by **2030**. Key SDGs: No Poverty (1), Zero Hunger (2), Good Health (3), Quality Education (4), Clean Water (6), Clean Energy (7), Climate Action (13), Life on Land (15).

Green Economy: Low-carbon, resource-efficient, socially inclusive economy. Features: Renewable energy, green jobs, sustainable agriculture, clean transportation, waste management.

Renewable vs Non-Renewable Energy

Aspect	Renewable Energy	Non-Renewable Energy
Sources	Solar, Wind, Hydro, Biomass	Coal, Petroleum, Natural Gas
Availability	Unlimited	Limited, will run out
Pollution	Low/No pollution	High pollution, greenhouse gases
Cost	High setup, low running	Lower setup, ongoing fuel costs

5.3 3 Rs, Carbon Footprint & Individual Responsibility

3 Rs: Reduce (use fewer resources), **Reuse** (use items again), **Recycle** (process waste into new products).

Carbon Footprint: Total greenhouse gases produced by a person or activity. Reduce by: saving energy, using public transport, eating locally, planting trees, reducing waste.

Climate Change: Long-term changes in temperature and weather caused by burning fossil fuels and deforestation. Effects: global warming, melting glaciers, rising sea levels, extreme weather.

Individual Role: Switch off lights, conserve water, avoid single-use plastics, plant trees, use public transport, spread awareness.

Key Environmental Days

World Environment Day: 5 June | Earth Day: 22 April | World Water Day: 22 March | World Ozone Day: 16 September

Keywords & MCQs

Sustainable Development | Brundtland Commission | SDGs | Green Economy | 3 Rs | Carbon Footprint | Renewable Energy | Climate Change | Green Jobs

1 Sustainable development was defined by:

a) WHO

b) Brundtland Commission

c) UNICEF

d) UNESCO

Answer: b) Brundtland Commission (1987).

2 The 3 Rs stand for:

a) Read, Review, Revise

b) Reduce, Reuse, Recycle

c) Repair, Replace, Remove

d) Retain, Return, Refresh

Answer: b) Reduce, Reuse, Recycle.

3 How many SDGs were set by the UN?

a) 10

b) 15

c) 17

d) 20

Answer: c) 17 SDGs adopted in 2015, to be achieved by 2030.

4 Which is a renewable source of energy?

a) Coal

b) Petroleum

c) Natural Gas

d) Solar Energy

Answer: d) Solar energy is renewable — unlimited and clean.

5 Carbon footprint measures:

a) Size of your foot

b) Total greenhouse gases produced by a person/activity

c) Amount of carbon in soil

d) Weight of CO_2 in atmosphere

Answer: b) Carbon footprint = total greenhouse gases produced by an individual or activity.

Unit 1

Introduction to Artificial Intelligence (AI)

Understand AI fundamentals — intelligence, AI domains, applications, and ethics.

PART B — Subject Specific Skills 15 Hours Theory 7 Marks

1.1 What is Intelligence?

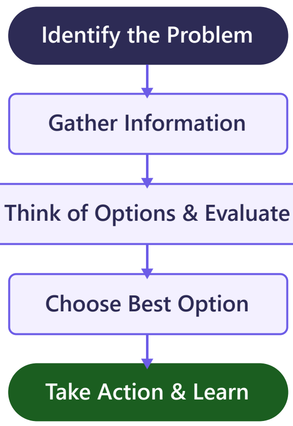
Definition

Intelligence is the ability to learn from experience, adapt to new situations, understand abstract concepts, reason logically, solve problems, and use knowledge effectively.

Components of Human Intelligence

Component	What It Means	Real-Life Example
Reasoning	Thinking logically and drawing conclusions	If cloudy and humid, you carry an umbrella
Problem-Solving	Finding solutions to complex issues	Fixing a broken circuit by testing each part
Creativity	Generating new, original ideas	Artist painting something never seen before
Learning	Acquiring knowledge through experience	Child learns not to touch hot stove after getting burned
Perception	Using senses to understand environment	Seeing red traffic light and stopping
Memory	Storing and recalling information	Remembering area formula during exam
Language	Understanding and producing communication	Reading Hindi newspaper and understanding news

Decision Making Process



Real-Life Example

Situation: Riya needs to reach school but it's raining heavily.
Options: Walk with umbrella, cycle with raincoat, ask father for car ride.
Decision: Car ride is safest → reaches school dry and on time.

1.2 What is Artificial Intelligence?

Simple Definition

AI is the ability of a machine or computer program to think, learn, and make decisions like a human being.

Technical Definition

AI is a branch of computer science that builds intelligent machines capable of performing tasks requiring human cognition — learning, reasoning, perception, language understanding, and problem solving.

Key Distinction

A system is AI if it can **learn**, **adapt**, and **improve** from experience. Simple automation following fixed rules is NOT AI.

AI vs NOT AI

Technology	AI or Not?	Why?
Google Assistant	**AI**	Understands language, learns preferences, adapts
Self-driving car (Tesla)	**AI**	Learns to recognize roads, pedestrians, adapts to new conditions
Gmail spam filter	**AI**	Learns from emails marked as spam, improves over time
Netflix recommendations	**AI**	Analyzes viewing history, suggests what you might like
Face unlock on phone	**AI**	Recognizes face even with glasses, haircut changes
Basic calculator	**Not AI**	Follows fixed math rules, cannot learn or adapt
Simple alarm clock	**Not AI**	Rings at set time, no learning
Fixed-timer traffic signal	**Not AI**	Changes on schedule regardless of traffic
Remote-controlled toy car	**Not AI**	No intelligence, follows human control only

Brief History of AI

Year	Event	Significance
1950	Alan Turing's paper	Proposed the Turing Test to check if machine can think
1956	Dartmouth Conference	Term "AI" coined by **John McCarthy** — "Father of AI"
1997	IBM Deep Blue beats Kasparov	Computer beats world chess champion
2011	Apple launches Siri	AI assistant reaches millions of users
2016	AlphaGo beats Go champion	AI masters the most complex board game
2022	ChatGPT launched	Generative AI accessible to public

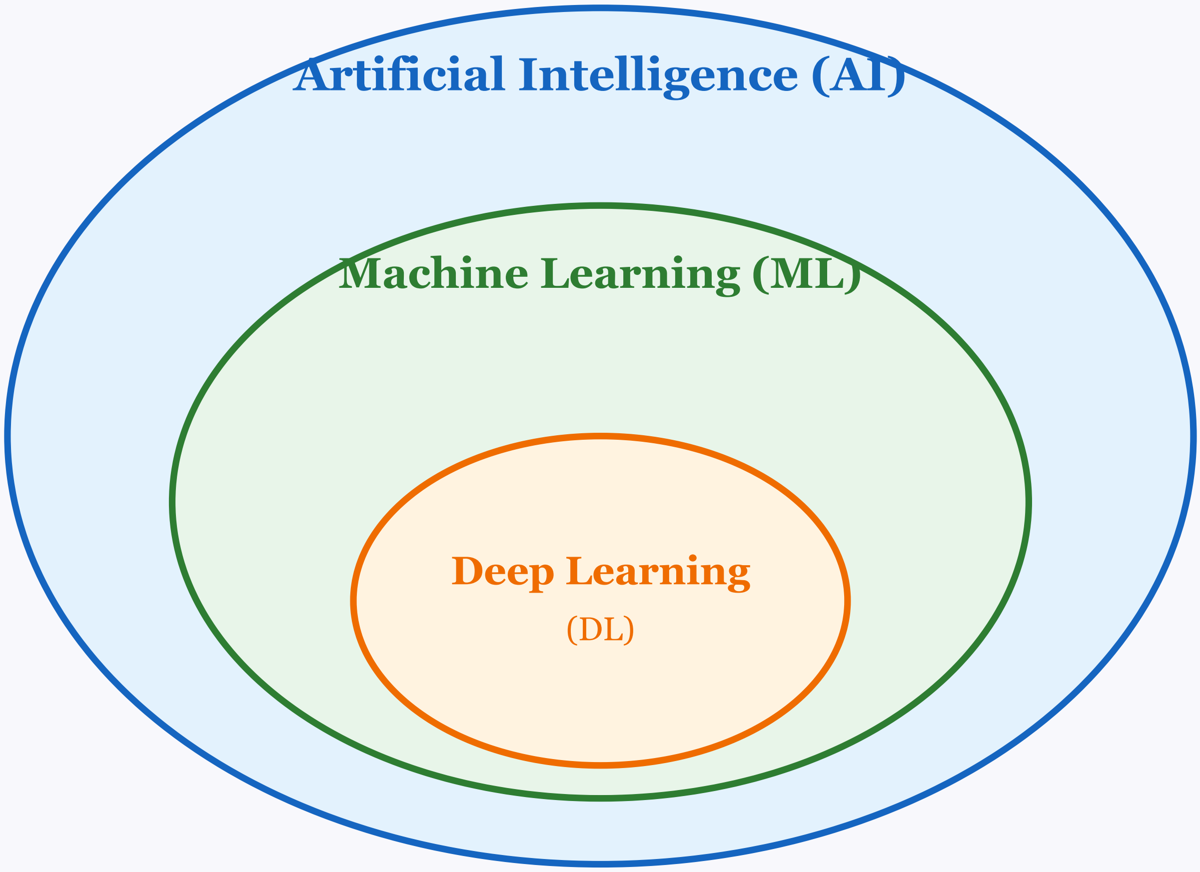
Exam Fact

John McCarthy coined "Artificial Intelligence" at the **Dartmouth Conference in 1956**. He is the "Father of AI."

1.3 AI vs ML vs DL

Definitions

AI: Broadest concept — any machine mimicking human intelligence.
ML (Machine Learning): Subset of AI — machines learn from data without explicit programming.
DL (Deep Learning): Subset of ML — multi-layered neural networks learning complex patterns.



AI ⊃ ML ⊃ DL — Nested relationship (DL inside ML inside AI)

Feature	AI	ML	DL
Scope	Broadest	Subset of AI	Subset of ML
How it works	Rules, logic, or learning	Learns from data	Neural networks, many layers
Data needed	Varies	Moderate	Very large amounts
Example	Chess computer	Spam filter	Face recognition, self-driving cars

Exam Tip

All DL is ML, all ML is AI, but not all AI is ML, and not all ML is DL.

Memory Trick — Russian Nesting Doll

AI = biggest doll | ML = middle doll inside AI | DL = smallest doll inside ML
Remember: **"All My Dogs"** → AI contains ML contains DL

1.4 Three Domains of AI

Domain	Deals With	What It Does	Examples	Tool
Data Science	Numbers & Data	Finds patterns & insights	Flipkart recommendations, weather forecasting	Pandas, Excel
Computer Vision (CV)	Images & Videos	Sees & understands visuals	Face unlock, Google Lens, self-driving cars	OpenCV, AutoDraw

NLP	Text & Speech	Understands human language	Google Translate, Alexa, chatbots	NLTK, Wordtune

Memory Trick — "DCN" for AI Domains

Data Science (Numbers) | **C**omputer Vision (Images) | **N**LP (Text & Speech)

1.5 Applications of AI

Industry	AI Application	Example
Healthcare	Disease detection from scans	AI reads chest X-rays for COVID/TB
Transportation	Self-driving, route optimization	Tesla, Google Maps traffic prediction
Finance	Fraud detection, credit scoring	UPI fraud detection system
Education	Personalized learning	BYJU'S adapts to student's speed
Agriculture	Crop monitoring, yield prediction	Plantix app identifies crop diseases
Entertainment	Content recommendations	Netflix, Spotify Discover Weekly
E-Commerce	Product recommendations	Amazon "Customers who bought this..."
Manufacturing	Quality control, robotics	Factory robots inspect for defects
Security	Surveillance, facial recognition	Airport security matching faces
Environment	Wildlife monitoring	AI cameras count endangered species

Indian AI Applications

India-Specific

UPI Fraud Detection: AI analyzes billions of transactions for unusual patterns.
Aadhaar Biometric: AI verifies identity via fingerprint/iris from 1.3 billion records.
IRCTC: AI predicts waitlist confirmation, detects booking bots.

1.6 AI Ethics

Definition

AI Ethics ensures AI is used fairly, safely, and responsibly — set of moral principles governing AI development and use.

AI Bias

AI Bias occurs when training data contains prejudices, leading to unfair results.

3 Examples of AI Bias

1. **Hiring:** AI trained on mostly male resumes penalized female applicants.
2. **Facial Recognition:** Higher error rates for darker skin tones due to imbalanced training data.
3. **Loan Approval:** AI rejected applications from certain neighborhoods regardless of individual creditworthiness.

Data Privacy

AI uses vast personal data (name, location, browsing, health records). Key concerns: Who collects? How stored? Who accesses? Can be misused? India's **Digital Personal Data Protection Act, 2023** protects citizens' data.

Moral Machine Experiment

MIT experiment: self-driving car's brakes fail — should it hit 3 elderly pedestrians or swerve into 1 child? Collected 40 million responses. Shows programming morality into machines is extremely complex.

Six Pillars of Responsible AI

Principle	What It Means	Violation Example

Fairness	No discrimination based on gender, race, etc.	AI rejecting loans based on gender
Transparency	People understand how AI decides	"Black box" AI denying insurance without explanation
Accountability	Someone responsible for AI actions	Self-driving car accident with no one held responsible
Safety	No harm to humans	Medical AI giving wrong diagnosis
Privacy	Protect personal data	App secretly recording conversations
Inclusiveness	Benefits everyone equally	Voice assistant can't understand regional accents

How to Address: Diverse training data, regular audits, explainable AI, human oversight, data protection laws, ethics committees.

Four Pillars Memory — "FTAP"

Fairness | **T**ransparency | **A**ccountability | **P**rivacy

1.7 Questions & Revision

AI | ML | DL | Data Science | Computer Vision | NLP | AI Bias | Data Privacy | Moral Machine | Ethics | John McCarthy | Dartmouth 1956 | Turing Test | Neural Network | Responsible AI

MCQs

1 Who coined "Artificial Intelligence"?

a) Alan Turing

b) John McCarthy

c) Charles Babbage

d) Tim Berners-Lee

Answer: b) John McCarthy, Dartmouth Conference, 1956.

2 Deep Learning is a subset of:

a) Data Science

b) NLP

c) Machine Learning

d) Computer Vision

Answer: c) DL is inside ML, which is inside AI.

3 Which is an example of AI?

a) Calculator

b) Alarm clock

c) Gmail spam filter that learns

d) Washing machine

Answer: c) Spam filter learns from data and adapts.

4 AutoDraw relates to which domain?

a) NLP

b) Data Science

c) Computer Vision

d) Robotics

Answer: c) AutoDraw uses CV to recognize sketches.

5 AI Bias occurs because:

a) AI is designed to discriminate

b) Training data contains biases

c) AI can't process data

d) AI is always fair

Answer: b) Biased data leads to biased AI outputs.

6 Google Translate is an example of:

a) CV

b) Data Science

c) NLP

d) Robotics

Answer: c) Translation involves understanding human language = NLP.

7 Correct relationship between AI, ML, DL:

a) AI inside ML inside DL

b) DL inside ML inside AI

c) ML inside DL inside AI

d) They're the same

Answer: b) DL ⊂ ML ⊂ AI.

8 "Birth of AI" event:

a) Computer invention 1940s

b) Dartmouth Conference 1956

c) Deep Blue 1997

d) ChatGPT 2022

Answer: b) Dartmouth Conference, 1956.

Short Answer Questions

Q1. Define AI. Give two examples. (2 marks)

Answer: AI is the ability of machines to think, learn, and make decisions like humans. Examples: (1) Google Assistant understanding voice commands. (2) Netflix recommending shows based on viewing history.

Q2. Name the three domains of AI with one example each. (3 marks)

Answer: (1) **Data Science** — analyzing data for patterns. Example: Weather forecasting. (2) **Computer Vision** — understanding images/videos. Example: Face unlock. (3) **NLP** — understanding language. Example: Google Translate.

Q3. What is AI Bias? Why is it a concern? (2 marks)

Answer: AI Bias = unfair results because training data contains prejudices. Concern: can discriminate in hiring, loans, facial recognition — causing real harm to people.

Q4. Differentiate AI and ML. (2 marks)

Answer: AI is the broader concept of machines performing intelligent tasks (may use rules). ML is a subset where machines learn from data without explicit programming. All ML is AI but not all AI is ML.

Q5. List four principles of Responsible AI. (2 marks)

Answer: Fairness (no discrimination), Transparency (explainable decisions), Accountability (someone responsible), Privacy (protect personal data).

Long Answer Questions

Q1. Explain AI, ML, DL with diagram and examples. (5 marks)

Answer: AI = broadest concept, any machine mimicking human intelligence (e.g., chatbot). **ML** = subset of AI, learns from data (e.g., spam filter). **DL** = subset of ML, uses neural networks with many layers (e.g., face recognition). Relationship: DL inside ML inside AI (nested circles). Key differences: AI may use rules; ML always learns from data; DL needs most data and uses neural networks.

Q2. Describe the three domains of AI with definitions, examples, and applications. (5 marks)

Answer: (1) **Data Science** — extracting insights from data. Example: IPL teams analyze player stats. Application: E-commerce recommendations. (2) **Computer Vision** — enabling machines to see images/videos. Example: Google Lens identifies objects. Application: Aadhaar iris scanning. (3) **NLP** — enabling machines to understand language. Example: Google Translate. Application: Voice assistants like Alexa.

Q3. Explain AI Ethics, AI Bias, and how to address ethical concerns. (5 marks)

Answer: AI Ethics = moral principles for fair, safe AI use. **AI Bias** = unfair results from biased training data. Examples: (1) Hiring AI penalized women, (2) Facial recognition errors for darker skin, (3) Loan AI rejected certain neighborhoods. **How to address:** Use diverse data, regular audits, explainable AI, human oversight, data protection laws (India's DPDP Act 2023), ethics committees.

Quick Revision — Unit 1

- Intelligence = learn, reason, solve, adapt, perceive
- AI = machines that think and learn like humans
- Father of AI: John McCarthy (Dartmouth, 1956)
- AI ⊃ ML ⊃ DL (nested)
- 3 Domains: Data Science (numbers), CV (images), NLP (language)
- AI Ethics pillars: Fairness, Transparency, Accountability, Privacy
- AI Bias = unfair results from biased training data
- Indian AI: UPI fraud detection, Aadhaar, IRCTC

Unit 2

AI Project Cycle

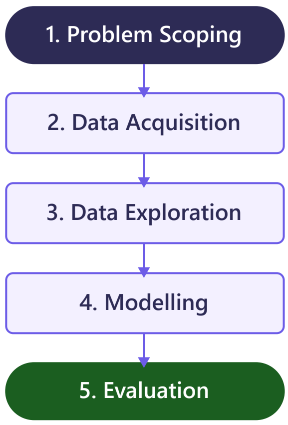
The systematic 5-stage framework for developing any AI solution — from problem to evaluation.

PART B — Subject Specific Skills   15 Hours Theory   9 Marks

2.1 The Five Stages

Definition

The **AI Project Cycle** is an iterative framework of five stages: Problem Scoping, Data Acquisition, Data Exploration, Modelling, and Evaluation.



Stage	What Happens	Key Question
1. Problem Scoping	Define the problem clearly	"What exactly is the problem?"
2. Data Acquisition	Collect needed data	"What data and where to get it?"
3. Data Exploration	Analyze and visualize data	"What does the data tell me?"
4. Modelling	Choose and train AI model	"Which AI technique to use?"
5. Evaluation	Test model performance	"How well does it work?"

Iterative Nature

The cycle is **iterative** — if evaluation shows poor results, go back to improve: need more data? → Data Acquisition. Wrong algorithm? → Modelling. Problem unclear? → Problem Scoping.

Memory Trick — "Please Don't Delay My Exam"

Problem Scoping → **D**ata Acquisition → **D**ata Exploration → **M**odelling → **E**valuation

2.2 Problem Scoping & 4Ws Canvas

W	Question	Example (Waste Management)	Example (Student Attendance)
WHO	Who has the problem?	Municipal corporation, residents	School, teachers, students
WHAT	What is the problem?	Garbage bins overflow	Manual attendance wastes time, proxy attendance
WHERE	Where does it occur?	Residential areas, markets	All classrooms, every period
WHY	Why solve it?	Health, reduce pollution	Track at-risk students early

SDG Connection

SDG	Goal	AI Project Connection
SDG 3	Good Health	AI for disease diagnosis
SDG 4	Quality Education	Personalized learning, attendance tracking
SDG 11	Sustainable Cities	Waste management, smart traffic
SDG 13	Climate Action	Weather prediction, deforestation monitoring

Writing a Problem Statement

Template: "Can we use [AI technique] to [solve what problem] for [whom] so that [desired outcome]?"

Example: "Can we use computer vision to detect crop diseases from leaf photos for farmers so they can treat plants early?"

2.3 Data Acquisition & Data Types

Sources: Surveys, Sensors/IoT, Web Scraping, Open Datasets (Kaggle, data.gov.in), Databases, APIs.

Category	Sub-type	Description	Example
Categorical	Nominal	No natural order	Colors, Gender, Blood Group
	Ordinal	Has natural order	Grades (A>B>C), Star ratings
Numerical	Discrete	Countable, whole numbers	Number of students: 35
	Continuous	Any value in range	Temperature: 37.5°C, Height: 165.4 cm

Data Formats: CSV (Comma Separated Values — most common), JSON (key-value pairs), SQL (relational databases).

2.4 Data Exploration & Visualization

Chart Type	Best Used For	Example
Bar Chart	Comparing categories	Marks of 5 students
Line Chart	Trends over time	Temperature over a week
Pie Chart	Proportions of a whole	Budget allocation
Scatter Plot	Relationship between 2 variables	Study hours vs marks
Histogram	Frequency distribution	Distribution of student heights

2.5 Modelling

Feature	Rule-Based	Learning-Based (ML)
How it works	Pre-defined IF-THEN rules	Learns patterns from data
Adaptability	Cannot adapt to new situations	Improves with more data
Example	Scripted chatbot	Spam filter that learns new spam

Types of Machine Learning

Type	Data	How It Works	Example	Analogy
Supervised	Labelled	Learns from input-output pairs	Email spam/not spam	Learning with teacher
Unsupervised	Unlabelled	Finds hidden patterns	Customer grouping	Learning without teacher
Reinforcement	Trial & error	Rewards & penalties	AI playing chess	Training a dog with treats

Activity — Unsupervised Learning Demo

Infinite Drum Machine (experiments.withgoogle.com/ai/drum-machine): Uses unsupervised learning to organise thousands of everyday sounds (not music) into groups based on similarity. The AI clusters similar-sounding items together without any labels — you can explore and create beats from the clusters. This demonstrates how unsupervised learning finds hidden patterns in unlabelled data.

Neural Networks

Inspired by human brain. Three layers: **Input Layer** (receives data) → **Hidden Layer(s)** (finds patterns) → **Output Layer** (produces prediction). Many hidden layers = Deep Learning.

2.6 Evaluation & Iteration

Tests model on new, unseen data. If evaluation shows poor results, **iterate**: go back to Data Acquisition (more data), Data Exploration (clean better), Modelling (try different algorithm), or even Problem Scoping (redefine problem). Detailed metrics in Unit 7.

2.7 Complete Project Example: Student Exam Prediction

Full AI Project Cycle Walkthrough

1. Problem Scoping (4Ws): WHO: School, teachers. WHAT: 15-20% students fail unexpectedly, no early warning. WHERE: Kendriya Vidyalaya, 400 students. WHY: Early identification helps teachers provide support. SDG 4.

2. Data Acquisition: 500 records from 3 years. Columns: Term 1 marks, attendance %, homework %, study hours, result (Pass/Fail). Sources: school database + Google Forms survey. Format: CSV.

3. Data Exploration: Average marks: 64.2 (Pass: 71.5, Fail: 41.3). Attendance: Pass group 86.2%, Fail group 58.4%. Scatter plot shows strong correlation between attendance and marks. Students with >85% attendance rarely fail.

4. Modelling: Supervised Learning (labelled data). Decision Tree Classifier. Split 80% training, 20% testing. Rules learned: "If Attendance<65% AND Marks<45 → Predict FAIL".

5. Evaluation: Accuracy: 87%. Correctly identified 22/25 who failed (88% recall). 3 missed (FN), 10 false alarms (FP — acceptable, extra attention doesn't hurt). Model ready for deployment as early warning system.

2.8 Questions & Revision

AI Project Cycle | Problem Scoping | 4Ws | Data Acquisition | Data Exploration | Modelling | Evaluation | Iterative | Supervised | Unsupervised | Reinforcement | Neural Network | Decision Tree | CSV | Nominal | Ordinal | Discrete | Continuous | SDG

MCQs

1 Correct order of AI Project Cycle:

a) Data → Problem → Model → Explore → Evaluate

b) Problem → Data Acquisition → Data Exploration → Modelling → Evaluation

c) Model → Explore → Problem → Data → Evaluate

Answer: b) Standard order. The cycle is iterative.

2 4Ws in Problem Canvas:

a) Who, What, When, Why

b) Who, What, Where, Why

c) Who, What, Where, Which

Answer: b) WHO, WHAT, WHERE, WHY.

3 Teachable Machine demonstrates:

a) Unsupervised

b) Reinforcement

c) Supervised

d) Rule-Based

Answer: c) Uses labelled examples = supervised learning.

4 "Blood Group (A,B,AB,O)" is which data type?

a) Discrete

b) Continuous

c) Nominal

d) Ordinal

Answer: c) Nominal — labels with no natural order.

5 "Height of students" is which type?

a) Nominal

b) Ordinal

c) Discrete

d) Continuous

Answer: d) Height can be any decimal value (152.5 cm).

6 Which ML type uses labelled data?

a) Unsupervised

b) Supervised

c) Reinforcement

d) None

Answer: b) Supervised uses labelled input-output pairs.

7 AI Project Cycle is "iterative" because:

a) Done only once

b) May go back to previous stages

c) Always linear

d) Only last stage repeats

Answer: b) Can go back to any stage to improve.

8 CSV stands for:

a) Computer Separated Values

b) Comma Separated Values

c) Character Separated Values

d) Code Separated Values

Answer: b) Comma Separated Values.

Short Answer Questions

Q1. What is the AI Project Cycle? Name its 5 stages. (2 marks)

Answer: A systematic 5-stage framework for AI solutions: (1) Problem Scoping, (2) Data Acquisition, (3) Data Exploration, (4) Modelling, (5) Evaluation. It is iterative — can go back to improve.

Q2. Explain the 4Ws Problem Canvas with an example. (3 marks)

Answer: Framework for defining problems: WHO (stakeholders), WHAT (problem), WHERE (location), WHY (importance). Example (Attendance): WHO: School, teachers. WHAT: Manual attendance wastes time, proxy attendance. WHERE: All classrooms. WHY: Track absenteeism early.

Q3. Differentiate Categorical and Numerical data. (2 marks)

Answer: Categorical: Labels/groups, no math. Nominal (no order: colors) and Ordinal (ordered: grades A>B). **Numerical:** Numbers for calculation. Discrete (whole: student count) and Continuous (any value: temperature 36.6°C).

Q4. Differentiate Supervised and Unsupervised Learning. (2 marks)

Answer: Supervised: Labelled data (input+output), learns to predict. Like learning with teacher. Example: spam filter. **Unsupervised:** Unlabelled data, discovers hidden patterns. Like learning alone. Example: customer grouping.

Q5. Why is the AI Project Cycle called "iterative"? (2 marks)

Answer: Because you can go back to any previous stage to improve: low accuracy → try different model (Modelling), data quality issues → collect more data (Data Acquisition), problem unclear → redefine (Problem Scoping).

Long Answer Questions

Q1. Explain all 5 stages of AI Project Cycle with an example. (5 marks)

Answer: Example: Crop Disease Detection. (1) **Problem Scoping:** WHO: Farmers. WHAT: Can't identify diseases early. WHY: Prevents crop loss. (2) **Data Acquisition:** Collect thousands of leaf photos labelled with diseases. Sources: Agricultural universities, PlantVillage dataset. (3) **Data Exploration:** Check image quality, ensure balanced data, observe patterns in diseased vs healthy leaves. (4) **Modelling:** Train CNN (supervised learning) on labelled images. (5) **Evaluation:** Test on new leaf images. If 90% accuracy, deploy as mobile app. If low, iterate — collect more data.

Q2. Explain the 3 types of ML with definitions, examples, and comparison. (5 marks)

Answer: (1) **Supervised:** Labelled data, learns input-output mapping. Like teacher giving answers. Example: spam filter trained on spam/not-spam emails. (2) **Unsupervised:** Unlabelled data, finds hidden groups. Like sorting objects without instructions. Example: customer segmentation on e-commerce. (3) **Reinforcement:** Agent learns by trial-and-error, rewards for good actions. Like training a dog with treats. Example: AlphaGo playing millions of Go games. Key difference: Supervised has labels, Unsupervised doesn't, Reinforcement uses rewards.

Quick Revision — Unit 2

- 5 stages: Problem Scoping → Data Acquisition → Data Exploration → Modelling → Evaluation
- Mnemonic: "Please Don't Delay My Exam"
- 4Ws: WHO, WHAT, WHERE, WHY
- Data: Categorical (Nominal, Ordinal) + Numerical (Discrete, Continuous)
- Formats: CSV, JSON, SQL
- Charts: Bar (compare), Line (trends), Pie (parts), Scatter (relationships), Histogram (distribution)
- Rule-Based = fixed rules | Learning-Based = learns from data

- Supervised = labelled | Unsupervised = unlabelled | Reinforcement = rewards
- Neural Network: Input → Hidden → Output layers
- Iterative = can go back to any stage to improve

Unit 3

Advance Python

Complete Python programming — variables, data types, operators, control flow, functions, lists, strings, dictionaries, NumPy, Pandas, Matplotlib.

PART B — Subject Specific Skills 30 Hours Practical Assessed in Practicals Only

3.1 What is Python?

Definition

Python is a high-level, interpreted, general-purpose programming language created by **Guido van Rossum in 1991**. It is known for its simple, readable syntax that is easy to learn.

Why Python for AI? (1) Simple syntax — reads like English. (2) Huge library ecosystem (NumPy, Pandas, TensorFlow). (3) Large community — help easily available. (4) Used by Google, Netflix, Instagram, NASA. (5) Free and open-source.

Feature	Python	C/C++/Java
Syntax	Simple, minimal code	Complex, more code needed
Type	Interpreted (runs line by line)	Compiled (whole program first)
Typing	Dynamic (no type declaration)	Static (must declare types)
AI Libraries	Excellent (NumPy, Pandas, TensorFlow)	Limited
Learning Curve	Easy for beginners	Harder to learn

3.2 Jupyter Notebook & Setup

Jupyter Notebook

A web-based interactive environment for writing and running Python code in cells. Install: `pip install jupyter`. Launch: `jupyter notebook`.

Cell types: Code Cell (Python code) and Markdown Cell (formatted text).

Shortcut	Action
Shift+Enter	Run current cell and move to next
Ctrl+Enter	Run current cell, stay in same cell
A	Insert cell above (in command mode)
B	Insert cell below
M	Change cell to Markdown
Y	Change cell to Code
DD	Delete cell (press D twice)
Ctrl+S	Save notebook

Virtual Environment & pip

Create: `python -m venv myenv`. Activate: `myenv\Scripts\activate` (Windows). Install packages: `pip install numpy pandas matplotlib opencv-python scipy`. List packages: `pip list`.

3.3 Variables & Data Types

Variable

> A **variable** is a named container that stores a value in memory. Think of it as a labelled box holding data.

Naming Rules: (1) Start with letter or underscore. (2) No spaces or special chars. (3) Case-sensitive (age ≠ Age). (4) Cannot use Python keywords (if, for, while). (5) Use meaningful names.

PYTHON — VARIABLES & DATA TYPES

```python
name = "Aarav"              # str (string)
age = 15                    # int (integer)
height = 5.6                # float (decimal)
is_student = True           # bool (True/False)
fruits = ["apple", "banana"] # list (mutable, ordered)
colors = ("red", "blue")    # tuple (immutable, ordered)
student = {"name": "Priya", "age": 15}  # dict (key-value pairs)
unique = {1, 2, 3}          # set (unique values, unordered)
print(type(age))            # <class 'int'>
```

All Data Types Comparison

Type	Example	Mutable?	Ordered?
int	10, -5, 0	No	-
float	3.14, -2.5	No	-
str	"hello", 'AI'	No	Yes
bool	True, False	No	-
list	[1, 2, 3]	**Yes**	Yes
tuple	(1, 2, 3)	No	Yes
dict	{"a": 1}	**Yes**	Yes*
set	{1, 2, 3}	**Yes**	No

> **Mutable vs Immutable**
>
> **Mutable** = can change after creation (list, dict, set). **Immutable** = cannot change (int, float, str, bool, tuple).

Type Conversion

PYTHON

```python
x = int("25")       # str to int: 25
y = float("3.14")   # str to float: 3.14
z = str(100)        # int to str: "100"
```

> **Important**
>
> `input()` ALWAYS returns a **string**. To do math with user input, convert: `age = int(input("Enter age: "))`.

3.3b Comments in Python

Definition

A **comment** is a line in the code that is not executed by Python. Comments are used to explain code, make it readable, and help others (or yourself) understand what the code does.

Types of Comments

Type	Syntax	Usage
Single-line Comment	`# comment text`	Explain one line or a short note
Multi-line Comment	`''' comment '''` or `""" comment """`	Explain a block of code or write docstrings

PYTHON – COMMENTS

```python
# This is a single-line comment

name = "Aarav"  # Variable to store student name

'''

This is a multi-line comment.

It can span multiple lines.

Used for longer explanations.

'''

def add(a, b):

    """This is a docstring. It describes what the function does."""

    return a + b
```

Why Use Comments?

- Make code easier to understand for yourself and others
- Explain complex logic or formulas
- Temporarily disable code during debugging ("commenting out")
- **Docstrings** (triple quotes inside functions) are special comments that describe what a function does — accessible via `help(function_name)`

Exam Tip

Comments are ignored by Python during execution. They do NOT affect the output. `#` is for single-line; `'''` or `"""` for multi-line. A docstring is the first string in a function body.

3.4 Operators

Type	Operators	Example	Result
Arithmetic	`+` Addition	`5 + 3`	8
	`-` Subtraction	`10 - 4`	6
	`*` Multiplication	`3 * 4`	12

	`/` Division (float)	`7 / 2`	3.5
	`//` Floor Division	`7 // 2`	**3**
	`%` Modulus (remainder)	`7 % 2`	**1**
	`**` Power	`2 ** 3`	**8**
Comparison	`== != > < >= <=`	`5 == 5`	True
Logical	`and or not`	`True and False`	False
Assignment	`= += -= *= /=`	`x += 5`	x = x + 5

Precedence (PEMDAS): Parentheses → Exponentiation (**) → Multiplication/Division → Addition/Subtraction.

Logical Operators Truth Table

A	B	A and B	A or B	not A
True	True	True	True	False
True	False	False	True	False
False	True	False	True	True
False	False	False	False	True

3.5 Input, Output & f-strings

PYTHON — PRINT() FEATURES

```python
print("Hello", "World")          # Hello World (default sep=" ")
print("Hello", "World", sep="-") # Hello-World
print("Hello", end=" ")             # No newline at end

name = "Priya"
print(f"Hello, {name}! You are {2025-2010} years old.")
# Output: Hello, Priya! You are 15 years old.
```

Complete Programs with Input

PROGRAM: SIMPLE CALCULATOR

```python
1 a = float(input("Enter first number: "))
2 b = float(input("Enter second number: "))
3 print(f"Sum = {a+b}, Difference = {a-b}, Product = {a*b}, Quotient = {a/b}")
```

Line 1-2: `input()` returns string, so `float()` converts to decimal number for math.

PROGRAM: AREA OF RECTANGLE

```python
length = float(input("Length: "))
width = float(input("Width: "))
area = length * width
```

```
print(f"Area = {area}")  # If 5, 3 entered: Area = 15.0
```

3.6 Conditional Statements

PROGRAM: GRADE CALCULATOR

```
marks = int(input("Enter marks: "))
if marks ≥ 90:
    print("Grade: A+")
elif marks ≥ 75:
    print("Grade: A")
elif marks ≥ 60:
    print("Grade: B")
elif marks ≥ 40:
    print("Grade: C")
else:
    print("Grade: F (Fail)")
```

PROGRAM: EVEN OR ODD

```
num = int(input("Enter number: "))
if num % 2 == 0:
    print(f"{num} is Even")
else:
    print(f"{num} is Odd")
```

PROGRAM: LARGEST OF THREE NUMBERS

```
a, b, c = 15, 27, 9
if a ≥ b and a ≥ c:
    print(f"Largest: {a}")
elif b ≥ c:
    print(f"Largest: {b}")  # Largest: 27
else:
    print(f"Largest: {c}")
```

3.7 Loops

FOR LOOP WITH RANGE()

```
for i in range(1, 6):
    print(i, end=" ")  # 1 2 3 4 5
```

`range(start, stop, step)` : stop is excluded. `range(1, 10, 2)` = 1, 3, 5, 7, 9.

WHILE LOOP

```
i = 1
while i ≤ 5:
    print(i, end=" ")  # 1 2 3 4 5
    i += 1
```

break = exit loop entirely. **continue** = skip this iteration. **pass** = do nothing (placeholder).

PROGRAM: MULTIPLICATION TABLE

```
num = int(input("Enter number: "))
for i in range(1, 11):
    print(f"{num} x {i} = {num*i}")
```

PROGRAM: FACTORIAL

```
n = int(input("Enter number: "))
fact = 1
for i in range(1, n+1):
    fact *= i
print(f"Factorial of {n} = {fact}")  # 5! = 120
```

PROGRAM: SUM OF N NUMBERS

```
n = int(input("Enter N: "))
total = 0
for i in range(1, n+1):
    total += i
print(f"Sum of 1 to {n} = {total}")  # N=10: Sum = 55
```

3.7b Functions (User-Defined)

Definition

> A **function** is a reusable block of code that performs a specific task. It runs only when it is **called**. Functions help in organizing code, avoiding repetition, and making programs modular.

Syntax

PYTHON – FUNCTION SYNTAX

```python
def function_name(parameter1, parameter2):    # Definition
    """Docstring: describes what function does"""
    # body of the function
    return result                                  # Optional

function_name(arg1, arg2)                      # Calling the function
```

Term	Meaning	Example
`def`	Keyword to define a function	`def greet():`
Parameters	Variables listed in function definition	`def add(a, b):` — a, b are parameters
Arguments	Values passed when calling	`add(5, 3)` — 5, 3 are arguments
`return`	Sends a value back to caller	`return a + b`

Programs Using Functions

PROGRAM 1: GREET FUNCTION

```python
def greet(name):
    print(f"Hello, {name}! Welcome to AI class.")

greet("Aarav")    # Hello, Aarav! Welcome to AI class.
greet("Priya")    # Hello, Priya! Welcome to AI class.
```

> The function `greet()` takes one parameter `name` and prints a personalised message. It is called twice with different arguments.

PROGRAM 2: ADD TWO NUMBERS (WITH RETURN)

```python
def add(a, b):
    return a + b

result = add(10, 25)
print(f"Sum = {result}")    # Sum = 35
```

> `return` sends the computed value back. Without `return`, the function returns `None` by default.

PROGRAM 3: CHECK EVEN OR ODD

```python
def check_even_odd(num):
```

```
    if num % 2 == 0:
        return "Even"
    else:
        return "Odd"

print(check_even_odd(7))    # Odd
print(check_even_odd(10))   # Even
```

PROGRAM 4: FACTORIAL USING FUNCTION

```
def factorial(n):
    fact = 1
    for i in range(1, n+1):
        fact *= i
    return fact

print(f"5! = {factorial(5)}")    # 5! = 120
print(f"7! = {factorial(7)}")    # 7! = 5040
```

PROGRAM 5: AREA OF CIRCLE (DEFAULT PARAMETER)

```
def area_circle(radius, pi=3.14159):  # pi has default value
    return pi * radius ** 2

print(f"Area = {area_circle(5)}")       # Area = 78.53975 (uses default pi)
print(f"Area = {area_circle(5, 3.14)}")  # Area = 78.5 (custom pi)
```

Default parameter: `pi=3.14159` means if no value is passed for pi, it uses 3.14159 automatically.

Scope of Variables

Type	Where Defined	Where Accessible	Example
Local Variable	Inside a function	Only inside that function	`x = 10` inside `def add():`
Global Variable	Outside all functions	Everywhere in the program	`name = "AI"` at top level

PYTHON – LOCAL VS GLOBAL SCOPE

```
x = 100             # Global variable

def my_func():
```

```
    y = 50          # Local variable (only inside my_func)
    print(x, y)     # Can access global x and local y

my_func()             # 100 50
print(x)              # 100 (global works here)
# print(y)            # ERROR! y is local to my_func
```

Built-in vs User-Defined Functions

Feature	Built-in Functions	User-Defined Functions
Source	Pre-defined in Python	Created by the programmer
Examples	`print()`, `len()`, `input()`, `type()`	`greet()`, `add()`, `factorial()`
Definition	No need to define	Must use `def` keyword
Availability	Always available	Available only after definition

Exam Tip

- `def` only **defines** a function — it does NOT run it. You must **call** the function to execute it.
- A function without `return` returns `None` by default.
- Parameters are in the definition; Arguments are in the call.
- Default parameters must come **after** non-default parameters.

Memory Trick — "DPCR"

Define (def) → **P**arameters (inputs) → **C**ode (body) → **R**eturn (output)

3.8 Lists

PYTHON – LIST OPERATIONS

```
fruits = ["apple", "banana", "cherry"]
print(fruits[0])      # apple (first)
print(fruits[-1])     # cherry (last)
print(fruits[0:2])    # ['apple', 'banana'] (slice)

fruits.append("date")     # Add at end
fruits.insert(1, "fig")   # Add at index 1
fruits.remove("banana") # Remove by value
fruits.pop()              # Remove last
fruits.sort()             # Sort ascending
fruits.reverse()          # Reverse order
print(len(fruits))       # Count items
```

```python

# List Comprehension
squares = [x**2 for x in range(1,6)]  # [1, 4, 9, 16, 25]
```

3.8b Tuples

Definition

A **tuple** is an **ordered, immutable** collection of items. Once created, you **cannot add, remove, or change** elements. Tuples are written with round brackets `()`.

PYTHON — TUPLE OPERATIONS

```python
# Creating Tuples
colors = ("red", "green", "blue")
single = (5,)              # Single element tuple NEEDS comma
mixed = (1, "hello", 3.14, True)

# Accessing Elements (same as list)
print(colors[0])        # red (first)
print(colors[-1])       # blue (last)
print(colors[0:2])      # ('red', 'green') (slicing)

# Tuple Operations
print(len(colors))      # 3 (length)
print("red" in colors)  # True (membership)
print(colors + ("yellow",))  # ('red', 'green', 'blue', 'yellow') (concatenation)
print(colors * 2)       # ('red', 'green', 'blue', 'red', 'green', 'blue') (repetition)

# Tuple Methods (only 2!)
nums = (1, 3, 5, 3, 7, 3)
print(nums.count(3))    # 3 (how many times 3 appears)
print(nums.index(5))    # 2 (first position of 5)
```

Tuples are Immutable!

`colors[0] = "pink"` → **TypeError!** You cannot change, add, or remove elements from a tuple after creation. To modify, create a new tuple.

Tuple vs List

Feature	Tuple	List
Syntax	`(1, 2, 3)`	`[1, 2, 3]`
Mutable?	**No** (cannot change)	**Yes** (can change)
Speed	Faster	Slower
Methods	Only 2: `count()`, `index()`	Many: `append()`, `remove()`, `sort()`, etc.
Use When	Data should NOT change (e.g., days of week, RGB values)	Data may change (e.g., shopping list, student marks)

Exam Tip

- Single-element tuple needs a comma: `(5,)` is a tuple, `(5)` is just an integer.
- Tuples have only 2 methods: `count()` and `index()`.
- Tuples are used as dictionary keys (lists cannot be).

3.8c Sets

Definition

A **set** is an **unordered, mutable** collection of **unique** elements. Sets automatically remove duplicates. Written with curly brackets `{}`.

PYTHON – SET OPERATIONS

```python
# Creating Sets
fruits = {"apple", "banana", "cherry"}
nums = {1, 2, 3, 2, 1}        # Stored as {1, 2, 3} — duplicates removed!

# Adding & Removing
fruits.add("date")            # Add one item
fruits.remove("banana")       # Remove (ERROR if not found)
fruits.discard("mango")       # Remove (NO error if not found)
fruits.pop()                  # Remove random element
fruits.clear()                # Remove ALL elements

# Set Properties
print(len(nums))             # 3 (unique count)
print(2 in nums)             # True (membership)
```

Mathematical Set Operations

PYTHON – SET MATH

```python
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}
```

```python

print(A | B)       # {1, 2, 3, 4, 5, 6}  — Union (all elements)
print(A & B)       # {3, 4}              — Intersection (common)
print(A - B)       # {1, 2}              — Difference (in A, not in B)
print(A ^ B)       # {1, 2, 5, 6}        — Symmetric Diff (not in both)
```

Operation	Operator	Method	Result
Union	`A | B`	`A.union(B)`	All elements from both sets
Intersection	`A & B`	`A.intersection(B)`	Common elements only
Difference	`A - B`	`A.difference(B)`	Elements in A but not in B
Symmetric Diff	`A ^ B`	`A.symmetric_difference(B)`	Elements in either, but not both

Real-Life Example

Remove duplicates from a list:

`names = ["Aarav", "Priya", "Aarav", "Riya", "Priya"]`
`unique_names = list(set(names))`
`print(unique_names) # ['Aarav', 'Priya', 'Riya']`

Exam Tip

- Sets are **unordered** — no indexing (`s[0]` gives ERROR).
- Sets store only **unique** values — duplicates are auto-removed.
- Use `discard()` instead of `remove()` to avoid errors when element doesn't exist.
- Empty set: `s = set()` (NOT `s = {}` which creates an empty dictionary!).

Memory Trick — List vs Tuple vs Set vs Dict

List = [square] = ordered, mutable, duplicates OK
Tuple = (round) = ordered, immutable, duplicates OK
Set = {curly} = unordered, mutable, NO duplicates
Dict = {curly: colon} = key-value pairs, keys unique

3.9 Strings

PYTHON — STRING METHODS

```python
s = "  Hello, World!  "
print(s.upper())      # "  HELLO, WORLD!  "
print(s.lower())      # "  hello, world!  "
print(s.strip())      # "Hello, World!" (remove spaces)
print(s.split(","))     # ['  Hello', ' World!  ']
print("-".join(["a","b","c"]))  # "a-b-c"
print(s.replace("World", "AI"))  # "  Hello, AI!  "
print(s.find("World"))  # 9 (index where found)
print(s.count("l"))     # 3
```

```python

# Indexing & Slicing
text = "PYTHON"
print(text[0])       # P
print(text[1:4])     # YTH
print(text[::-1])    # NOHTYP (reversed)
```

Strings are Immutable
You cannot change a character in a string. `s[0] = "h"` gives an error. String methods return NEW strings.

3.10 Dictionaries

PYTHON — DICTIONARY OPERATIONS

```python
student = {"name": "Aarav", "age": 15, "grade": "A"}

print(student["name"])       # Aarav
print(student.get("age"))    # 15 (safe access)
student["marks"] = 92        # Add new key
student["age"] = 16          # Update existing
del student["grade"]         # Delete key

print(student.keys())    # dict_keys(['name', 'age', 'marks'])
print(student.values())  # dict_values(['Aarav', 16, 92])
print(student.items())   # key-value pairs

for key, val in student.items():
    print(f"{key}: {val}")
```

3.11 Built-in Functions & Libraries

Function	Purpose	Example	Output
`len()`	Count items	`len([1,2,3])`	3
`type()`	Check data type	`type(3.14)`	<class 'float'>
`max()`	Largest value	`max(3,7,1)`	7
`min()`	Smallest value	`min(3,7,1)`	1
`sum()`	Add all items	`sum([1,2,3])`	6

`sorted()`	Sort (new list)	`sorted([3,1,2])`	[1,2,3]
`abs()`	Absolute value	`abs(-5)`	5
`round()`	Round number	`round(3.7)`	4

Importing Libraries

PYTHON

```python
import math
print(math.sqrt(25))   # 5.0
print(math.pi)         # 3.14159...

import random
print(random.randint(1, 10))  # Random 1-10
print(random.choice(["a","b","c"]))  # Random pick
```

3.12 NumPy, Pandas, Matplotlib Basics

Detailed coverage is in Unit 4 (Data Science). Quick reference here:

NUMPY QUICK REFERENCE

```python
import numpy as np
a = np.array([10,20,30]); print(np.mean(a), np.median(a), np.std(a))
```

PANDAS QUICK REFERENCE

```python
import pandas as pd
df = pd.read_csv("data.csv"); print(df.head(), df.shape, df.describe())
```

MATPLOTLIB QUICK REFERENCE

```python
import matplotlib.pyplot as plt
plt.plot([1,2,3],[4,5,6]); plt.title("Chart"); plt.show()
```

3.13 Common Errors & Debugging

Error	Cause	Example	Fix
SyntaxError	Wrong syntax	Missing colon after if	Check colons, brackets, quotes
NameError	Variable not defined	Using `nam` instead of `name`	Check spelling
TypeError	Wrong data type	`"5" + 3` (str + int)	Convert: `int("5") + 3`
IndexError	Index out of range	`a[10]` when list has 5 items	Check list length first
ValueError	Wrong value	`int("hello")`	Ensure convertible input

IndentationError	Wrong spacing	Mixed tabs and spaces	Use consistent 4-space indent
ZeroDivisionError	Dividing by zero	`10 / 0`	Check divisor before dividing

3.14 Questions & Revision

Python | Jupyter | Variable | int | float | str | bool | list | tuple | dict | set | Mutable | Immutable | Operators | if-elif-else | for loop | while loop | range() | break | continue | input() | print() | f-string | Slicing | append() | sort() | split() | join() | import | math | random | NumPy | Pandas | Matplotlib | pip | PEMDAS

MCQs

1 `input()` always returns:

a) Integer

b) Float

c) String

d) Boolean

Answer: c) Always returns string. Convert with int() or float().

2 Output of `7 // 2` ?

a) 3.5

b) 3

c) 4

d) 3.0

Answer: b) Floor division rounds down. 7//2 = 3.

3 Which is immutable?

a) List

b) Dictionary

c) Tuple

d) Set

Answer: c) Tuples cannot be changed after creation.

4 Who created Python?

a) Dennis Ritchie

b) James Gosling

c) Guido van Rossum

d) Bjarne Stroustrup

Answer: c) Guido van Rossum, 1991.

5 Which is mutable?

a) int

b) str

c) tuple

d) list

Answer: d) Lists can be changed after creation.

6 `len([10, 20, 30, 40])` returns:

a) 100

b) 40

c) 4

d) 3

Answer: c) 4 items in the list.

7 Which keyword exits a loop?

a) continue

b) pass

c) break

d) exit

Answer: c) break exits the loop entirely.

8 `type(3.14)` returns:

a) int

b) float

c) str

d) decimal

Answer: b) 3.14 has decimal = float.

9 Which reads CSV files?

a) pd.load_csv()

b) pd.open_csv()

c) pd.read_csv()

d) pd.import_csv()

Answer: c) pd.read_csv("file.csv").

10 Correct way to create a dictionary:

a) d = [1,2,3]

b) d = (1,2,3)

c) d = {"key":"value"}

d) d = {1,2,3}

Answer: c) Curly braces with key:value pairs. (d) is a set.

Output Prediction

1 **Output of:** `x=5; y=2; print(x**y, x%y, x//y)`

Answer: 25 1 2 — 5^2=25, 5%2=1(remainder), 5//2=2(floor).

2 **Output of:** `fruits=["apple","banana","cherry"]; fruits.append("date"); print(len(fruits), fruits[-1])`

Answer: 4 date — After append, 4 items. fruits[-1] = last = "date".

3 **Output of:** `for i in range(1,6): if i==3: continue; print(i, end=" ")`

Answer: 1 2 4 5 — continue skips 3, prints rest.

4 **Output of:** `text="PYTHON"; print(text[1:4], text[::-1])`

Answer: YTH NOHTYP — [1:4]=index 1,2,3. [::-1]=reversed.

5 **Output of:** `d={"a":1,"b":2,"c":3}; d["b"]=20; print(d["b"], len(d))`

Answer: 20 3 — Updated b to 20. Still 3 keys.

Quick Revision — Python

- Created by Guido van Rossum, 1991 | Interpreted, dynamic typing
- input() always returns STRING — convert with int()/float()
- Mutable: list, dict, set | Immutable: int, float, str, bool, tuple
- / = float division | // = floor | % = remainder | ** = power
- for = fixed iterations | while = condition-based
- break = exit loop | continue = skip iteration | pass = do nothing
- List: append(), remove(), pop(), sort(), reverse()
- String: upper(), lower(), strip(), split(), join(), replace()
- Dict: keys(), values(), items(), get()
- f-string: f"Hello {name}" for formatting
- NumPy: arrays, mean(), median() | Pandas: DataFrames, read_csv()
- Matplotlib: plot(), bar(), scatter(), pie(), show()
- Always call plt.show() for charts

Unit 4

Data Science

Collecting, cleaning, exploring, and visualizing data — NumPy, Pandas, Matplotlib with Python.

PART B — Subject Specific Skills   7 hrs theory + 8 hrs practical   4 Marks Theory

4.1 What is Data Science?

Simple Definition

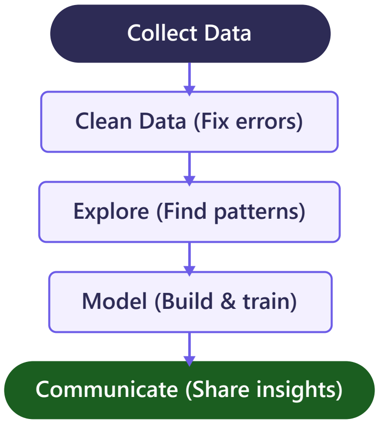
Data Science is the process of collecting, organising, analysing data, and finding useful patterns to make better decisions.

Technical Definition

Data Science is an interdisciplinary field that uses scientific methods, algorithms, and systems to extract knowledge from structured and unstructured data. It combines statistics, programming, and domain expertise.

Why data is important in AI: Data is the **"fuel of AI"**. Without data, AI cannot learn. More quality data = better predictions.

Data Science Workflow



Real-Life Examples

Industry	Application	Example
E-Commerce	Product recommendations	Amazon/Flipkart "Recommended for you"
Entertainment	Content recommendations	Netflix suggests shows, Spotify "Discover Weekly"
Sports	Player analytics	IPL teams select players using stats
Healthcare	Disease prediction	Predicting diabetes from patient records
Finance	Fraud detection	Detecting unusual credit card transactions
Transport	Route optimization	Google Maps real-time traffic navigation
Agriculture	Crop yield prediction	Predicting best time to sow seeds
Education	Student performance	Identifying at-risk students early

Workflow: "Can Cats Eat Many Cookies?"

Collect → **C**lean → **E**xplore → **M**odel → **C**ommunicate

4.2 Data Collection

Methods: Surveys/Questionnaires (Google Forms), Sensors/IoT (temperature, fitness trackers), Web Scraping (extracting from websites), Open Datasets (Kaggle, data.gov.in), APIs (weather data), Databases (school records).

Primary vs Secondary Data

Feature	Primary Data	Secondary Data
Definition	Collected first-hand for specific purpose	Already collected by someone else
Source	Surveys, experiments	Books, websites, government data
Cost	Expensive and time-consuming	Cheaper and faster
Example	Surveying students in your school	Using census data from data.gov.in

Structured vs Unstructured Data

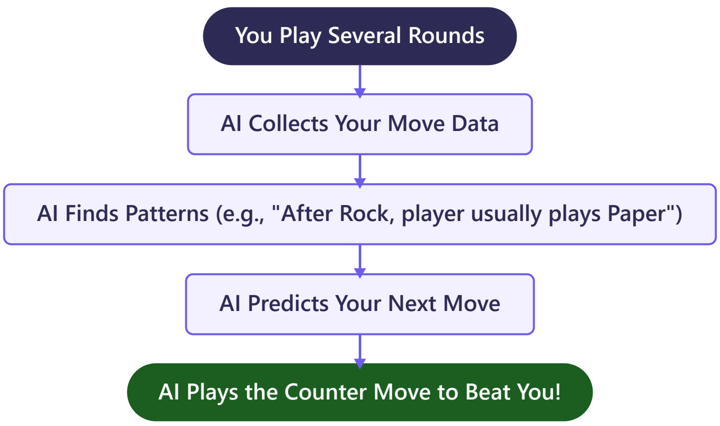
Feature	Structured	Unstructured
Format	Rows and columns (tables)	No fixed format
Examples	Student marks table, Excel, CSV	Images, videos, emails, social media posts
Easy to analyse?	Yes	Difficult, needs special tools

4.2b Activity: Rock, Paper & Scissors AI

What is This Activity?

Rock, Paper & Scissors AI is an interactive game where you play the classic Rock-Paper-Scissors game against an AI opponent. The AI **collects data** from your moves, **detects patterns** in your choices, and **predicts your next move** to beat you — demonstrating how data science works in real time.

How the AI Learns (Step by Step)



How to Play

Step	What to Do	What to Observe
1	Open the game (search "Rock Paper Scissors AI" or visit `next.rockpaperscissors.ai`)	The AI starts with random moves (no data yet)
2	Play 10-15 rounds with a **fixed pattern** (e.g., Rock-Paper-Scissors-Rock-Paper-Scissors...)	The AI starts **winning more** as it detects your pattern
3	Now play **completely randomly**	The AI struggles — random choices have no pattern to learn
4	Try playing with a **slight bias** (e.g., choosing Rock 60% of the time)	The AI detects your bias and plays Paper more often

Data Science Concepts Demonstrated

Concept	How It Appears in the Game
Data Collection	AI records every move you make (your choices = data)
Data Exploration	AI analyses your move history to find patterns and frequencies
Pattern Recognition	AI finds trends like "player picks Rock after losing" or "player repeats winning moves"
Prediction (Modelling)	AI uses patterns to predict your next move
Evaluation	AI checks if its predictions were correct and adjusts strategy
Iterative Improvement	The more you play, the more data AI has, and the better it gets

Key Takeaway

• **More data = Better predictions.** After 5 rounds, AI is guessing. After 50 rounds, AI is winning consistently.
• **Patterns make you predictable.** Humans are bad at being truly random — AI exploits this.
• **Random = Unbeatable.** If you play truly randomly, no AI can predict your move (but humans rarely achieve true randomness).

DIY: Build Your Own Rock-Paper-Scissors AI

PYTHON — SIMPLE RPS AI

```
1  import random
2
3  history = []  # Data collection: store player's moves
4  counter = {"rock": "paper", "paper": "scissors", "scissors": "rock"}
5
6  for _ in range(10):  # Play 10 rounds
7      player = input("Rock/Paper/Scissors: ").lower()
8      if len(history) < 3:
9          ai = random.choice(["rock", "paper", "scissors"])  # Random at start
10     else:
11         # Predict: player's most common move, then counter it
12         most_common = max(set(history), key=history.count)
13         ai = counter[most_common]  # Play the counter move
14     history.append(player)  # Store data for future predictions
15     print(f"AI played: {ai}")
```

Lines 3-4: Data collection (history list) and counter strategy. **Lines 8-13:** AI plays randomly for first 3 rounds (not enough data), then predicts using the most frequent move. **Line 14:** Every move is stored — more data = better prediction.

Exam Question Connection

This activity directly demonstrates the **AI Project Cycle** in action: Problem Scoping (beat the player) → Data Acquisition (record moves) → Data Exploration (find patterns) → Modelling (predict next move) → Evaluation (did AI win?).

4.3 Data Access & CSV Files

CSV (Comma Separated Values)
A simple text file where each line is a data record and fields are separated by commas. Most widely used data format.

CSV FILE — STUDENTS.CSV

```
Name,Maths,Science,English
Aarav,85,90,78
Priya,92,88,95
Rohan,76,82,80
```

Reading CSV with Pandas

PYTHON

```python
1 import pandas as pd
2 df = pd.read_csv("students.csv")
3 print(df)
```

Line 2: `pd.read_csv()` reads CSV into a DataFrame (table in Python). Index starts at 0.

4.4 Data Cleaning

Definition
Data Cleaning is fixing errors, missing values, duplicates, and inconsistencies. Takes 60-80% of total effort. **"Garbage In, Garbage Out"** — dirty data = incorrect results.

Common Issues & Solutions

Issue	Example	Solution
Missing Values	Student's Science marks empty	Fill with mean `df.fillna(df.mean())` or drop row `df.dropna()`
Duplicates	Same student entered twice	`df.drop_duplicates()`
Inconsistent Formats	"Delhi", "delhi", "DELHI"	Standardise to one format (lowercase)
Outliers	Marks = 950 (likely typo for 95)	Investigate and correct

PYTHON — HANDLING MISSING VALUES

```python
import pandas as pd
import numpy as np
data = {"Name": ["Aarav", "Priya", "Rohan"], "Maths": [85, np.nan, 76]}
df = pd.DataFrame(data)
print(df.isnull().sum())  # Shows missing count per column
df["Maths"] = df["Maths"].fillna(df["Maths"].mean())  # Fill with average
```

4.5 Data Visualization

Definition
Graphical representation of data using charts. Humans understand pictures faster than numbers.

Chart	Best For	Example
Bar Chart	Comparing categories	Marks of 5 students
Line Chart	Trends over time	Temperature over a week
Pie Chart	Parts of a whole (%)	Grade distribution
Scatter Plot	Relationship between 2 variables	Study hours vs marks
Histogram	Frequency distribution	How many scored 40-60, 60-80, etc.

Code Examples

PYTHON – BAR CHART

```python
import matplotlib.pyplot as plt
plt.bar(["Maths", "Sci", "Eng"], [85, 90, 78], color="skyblue")
plt.title("Subject-wise Marks"); plt.xlabel("Subject"); plt.ylabel("Marks")
plt.show()
```

PYTHON – LINE CHART

```python
import matplotlib.pyplot as plt
plt.plot(["Mon","Tue","Wed","Thu","Fri"], [32,35,33,36,34], marker="o", color="red")
plt.title("Weekly Temperature"); plt.grid(True); plt.show()
```

PYTHON – SCATTER PLOT

```python
import matplotlib.pyplot as plt
plt.scatter([1,2,3,4,5,6,7,8], [35,45,50,60,72,78,85,92], color="purple")
plt.xlabel("Hours Studied"); plt.ylabel("Marks"); plt.title("Study vs Marks"); plt.show()
```

PYTHON – PIE CHART

```python
import matplotlib.pyplot as plt
plt.pie([35,25,20,20], labels=["Math","Sci","AI","Eng"], autopct="%1.1f%%")
plt.title("Subject Distribution"); plt.show()
```

PYTHON – HISTOGRAM

```python
import matplotlib.pyplot as plt
plt.hist([45,67,78,85,92,55,73,88], bins=4, color="orange", edgecolor="black")
```

```python
plt.xlabel("Marks Range"); plt.ylabel("Count"); plt.title("Marks Distribution"); plt.show()
```

4.6 Statistical Concepts

Mean (Average)

Formula

Mean = Sum of all values ÷ Number of values

Solved Example

Marks: 78, 85, 92, 88, 77. Mean = (78+85+92+88+77)/5 = 420/5 = **84**

PYTHON

```python
import numpy as np
print(np.mean([78, 85, 92, 88, 77]))  # 84.0
```

Median (Middle Value)

How to Find

Sort data. Odd count: middle value. Even count: average of two middle values.

Solved (Odd)

12, 5, 8, 3, 15 → Sorted: 3, 5, **8**, 12, 15 → Median = **8**

Solved (Even)

10, 20, 30, 40 → Two middle: 20, 30 → Median = (20+30)/2 = **25**

Mode (Most Frequent)

Example

5, 3, 7, 3, 8, 3, 9 → **3** appears 3 times = Mode

PYTHON

```python
from scipy import stats
print(stats.mode([5,3,7,3,8,3,9]).mode)  # 3
```

Standard Deviation

Definition

Measures how spread out values are from the mean. Low SD = consistent. High SD = spread out.

Example

Class A: 78, 80, 82 → Mean=80, SD≈1.4 (consistent)
Class B: 50, 95, 40, 100, 65 → Mean=70, SD≈23 (spread out)

When to Use Which?

Measure	Use When	Sensitive to Outliers?
Mean	Data evenly distributed, no extremes	Yes
Median	Data has outliers or is skewed	No

Mode	Categorical data, most common value	No
Std Dev	Measuring consistency/spread	Yes

Solved Numerical Problems

Problem 1

Data: 4, 7, 2, 9, 4, 6, 4, 8. Find Mean, Median, Mode.
Mean = 44/8 = **5.5**. Sorted: 2,4,4,4,6,7,8,9. Median = (4+6)/2 = **5**. Mode = **4** (appears 3 times).

Problem 2

Salaries (thousands): 20, 25, 22, 24, 200. Mean = 291/5 = **58.2**. Median (sorted: 20,22,24,25,200) = **24**.
Median is better here because mean is distorted by outlier 200.

Mean, Median, Mode Memory Trick

Mean = "Mean is the Machine" (calculates: add & divide)
Median = "Median is in the Middle" (both start M-E-D)
Mode = "Mode is the Most" (both start M-O)

4.7 NumPy for Data Science

What is NumPy?

NumPy (Numerical Python) — library for arrays and math operations. Much faster than Python lists.

PYTHON – NUMPY ESSENTIALS

```python
import numpy as np

a = np.array([10, 20, 30, 40])
z = np.zeros(5)        # [0. 0. 0. 0. 0.]
o = np.ones(4)         # [1. 1. 1. 1.]
r = np.arange(0,10,2)  # [0 2 4 6 8]
l = np.linspace(0,1,5) # [0. 0.25 0.5 0.75 1.]

# Element-wise operations
b = np.array([1, 2, 3, 4])
print(a + b)    # [11 22 33 44]
print(a * 2)    # [20 40 60 80]

# Statistics
marks = np.array([85, 90, 78, 92, 88])
print(np.mean(marks))     # 86.6
print(np.median(marks))   # 88.0
```

```python
print(np.std(marks))     # 4.78
print(np.min(marks), np.max(marks), np.sum(marks))  # 78 92 433
print(marks.shape, marks.dtype, marks.size)  # (5,) int64 5
```

4.8 Pandas for Data Science

> **What is Pandas?**
> Library for data manipulation. Two structures: **Series** (1D column) and **DataFrame** (2D table).

PYTHON — PANDAS ESSENTIALS

```python
import pandas as pd

# Creating DataFrame from dict
data = {"Name": ["Aarav", "Priya"], "Marks": [85, 92]}
df = pd.DataFrame(data)

# Reading CSV
df = pd.read_csv("data.csv")
print(df.head(10))    # First 10 rows
print(df.tail(5))     # Last 5 rows
print(df.shape)       # (rows, cols)
print(df.columns)     # Column names
print(df.dtypes)      # Data types
print(df.describe())  # Statistics: count,mean,std,min,25%,50%,75%,max
df.info()              # Column types, non-null counts

# Filtering & Selecting
print(df["Name"])            # Single column
toppers = df[df["Marks"] > 80] # Filter rows
print(df.sort_values("Marks", ascending=False))  # Sort descending
```

> **loc** = label-based (uses column names). **iloc** = integer position-based (uses 0,1,2...).

4.9 Matplotlib Functions Summary

Function	Purpose
`plt.plot()`	Line chart

`plt.bar()`	Bar chart
`plt.scatter()`	Scatter plot
`plt.pie()`	Pie chart
`plt.hist()`	Histogram
`plt.xlabel()` / `plt.ylabel()`	Axis labels
`plt.title()`	Chart title
`plt.legend()`	Show legend
`plt.grid(True)`	Grid lines
`plt.show()`	Display chart — **MUST NOT forget!**

4.9b K-Nearest Neighbour (KNN) Model

Optional — Practical Only

Note for Students

This topic is **Optional** and is intended for practical assessments/learning only. It will not be assessed in the written board theory exam.

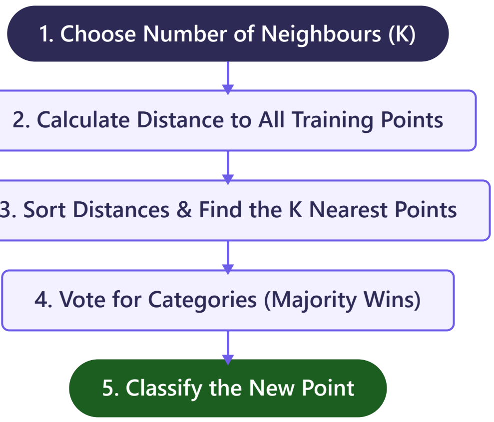
What is KNN?

K-Nearest Neighbour (KNN) is a simple, easy-to-implement **Supervised Machine Learning** algorithm used for both classification (sorting into groups) and regression (predicting values). It classifies a new data point based on how close it is to its neighboring data points.

The Core Idea: "Birds of a feather flock together"

If you want to know what category a new point belongs to, look at its **K closest neighbors**. Whichever category has the majority vote among those neighbors is the category assigned to the new point.

How KNN Works (Step-by-Step)



Key Concepts

Concept	Meaning	Explanation
Distance Metric	How to measure closeness	Usually **Euclidean Distance** (straight-line distance between two points: $d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$).
Value of K	Number of neighbours to check	If `K = 3`, we look at the 3 closest neighbors. Always choose an **odd number** for K to avoid tie votes.
Classification	Predicting a label (discrete)	e.g., Is this email "Spam" or "Not Spam"?

Activity: Personality Prediction (Introvert vs Extrovert)

In this activity, we use a simple 2D dataset to classify a person's personality type based on two traits:

1. **Social Energy (1-10):** How comfortable they are talking to strangers/large crowds.
2. **Focus Preference (1-10):** How much they prefer working alone in a quiet room.

Step 1: Training Data (Existing Profile Records)

Person	Social Energy (X)	Quiet Focus (Y)	Personality Label
Aman	8	2	**Extrovert**
Bina	2	9	**Introvert**
Charu	9	3	**Extrovert**
Dev	3	8	**Introvert**
Esha	4	7	**Introvert**

Step 2: The Query (Predict for a New Person)

Farhan has **Social Energy** = **7** and **Quiet Focus** = **4**. Is he an Introvert or Extrovert?

Step 3: Calculating Distance to Farhan (7, 4)

- Distance to Aman (8, 2) = $\sqrt{(8-7)^2 + (2-4)^2} = \sqrt{1 + 4} = \sqrt{5} \approx 2.24$
- Distance to Bina (2, 9) = $\sqrt{(2-7)^2 + (9-4)^2} = \sqrt{25 + 25} = \sqrt{50} \approx 7.07$
- Distance to Charu (9, 3) = $\sqrt{(9-7)^2 + (3-4)^2} = \sqrt{4 + 1} = \sqrt{5} \approx 2.24$
- Distance to Dev (3, 8) = $\sqrt{(3-7)^2 + (8-4)^2} = \sqrt{16 + 16} = \sqrt{32} \approx 5.66$
- Distance to Esha (4, 7) = $\sqrt{(4-7)^2 + (7-4)^2} = \sqrt{9 + 9} = \sqrt{18} \approx 4.24$

Step 4: Find the K Nearest Neighbours (Let K = 3)

Sorting distances from smallest to largest:

1. **Aman (2.24) → Extrovert**
2. **Charu (2.24) → Extrovert**
3. **Esha (4.24) → Introvert**

Step 5: Voting & Classifying

Among Farhan's 3 nearest neighbors, we have **2 Extroverts** (Aman, Charu) and **1 Introvert** (Esha).
Since Extrovert has the majority vote (2 out of 3), **Farhan is classified as an Extrovert!**

DIY: Python Code for KNN (No External ML Libraries Required)

PYTHON - DIY KNN

```python
import math

# Training dataset: [Social Energy, Quiet Focus, Label]
dataset = [
    [8, 2, "Extrovert"],
    [2, 9, "Introvert"],
    [9, 3, "Extrovert"],
    [3, 8, "Introvert"],
    [4, 7, "Introvert"]
]

```

```python
def euclidean_distance(point1, point2):
    return math.sqrt((point1[0] - point2[0])**2 + (point1[1] - point2[1])**2)

def predict_personality(new_person, k=3):
    distances = []
    for data in dataset:
        dist = euclidean_distance(new_person, data)
        distances.append((dist, data[2]))  # Store (distance, label)

    # Sort by distance (first item in tuple)
    distances.sort()
    neighbors = distances[:k]  # Take first K neighbors

    # Count votes
    votes = [label for dist, label in neighbors]
    return max(set(votes), key=votes.count)  # Return majority vote

farhan = [7, 4]
prediction = predict_personality(farhan, k=3)
print(f"Farhan's predicted personality: {prediction}")  # Extrovert
```

Why choose odd K?
If you choose an even number like `K = 2` or `K = 4`, there is a risk of a tie (e.g. 1 Extrovert vote and 1 Introvert vote). An odd number ensures a clear tie-breaker majority vote.

4.10 Questions & Revision

Data Science | CSV | DataFrame | Mean | Median | Mode | Standard Deviation | NumPy | Pandas | Matplotlib | Bar Chart | Line Chart | Pie Chart | Scatter | Histogram | Primary Data | Secondary Data | NaN | fillna | dropna | read_csv | loc/iloc | describe() | Outlier

MCQs

1 Which function displays first 5 rows?

a) df.tail()

b) df.info()

c) df.head()

d) df.describe()

Answer: c) head() returns first 5 rows by default.

2 Median of 12, 15, 18, 22, 25?

a) 15

b) 18

c) 18.4

d) 22

Answer: b) 5 values (odd), middle = 3rd = 18.

3 Which chart shows trends over time?

a) Pie

b) Bar

c) Line

d) Scatter

Answer: c) Line charts show trends over time.

4 Which library uses DataFrames?

a) NumPy

b) Matplotlib

c) Pandas

d) OpenCV

Answer: c) Pandas provides DataFrames.

5 Which reads CSV files?

a) pd.load_csv()

b) pd.open_csv()

c) pd.read_csv()

d) pd.import_csv()

Answer: c) pd.read_csv("file.csv").

6 NaN stands for:

a) Not a Name

b) Not a Number

c) Null and None

d) New and Null

Answer: b) NaN = Not a Number (missing value).

7 Which measure is NOT affected by outliers?

a) Mean

b) Median

c) Std Dev

d) Sum

Answer: b) Median is the middle value, unaffected by extremes.

8 **df.shape returns:**

a) Column names

b) Data types

c) (rows, columns) tuple

d) First 5 rows

Answer: c) e.g., (50, 4) = 50 rows, 4 columns.

Short & Long Answer Questions

Q1. What is data science? Give two examples. (2 marks)

Answer: Data science extracts useful information from data using statistics and programming. Examples: (1) Netflix recommends movies from viewing history. (2) Weather forecasting from temperature/humidity data.

Q2. Differentiate structured and unstructured data. (2 marks)

Answer: Structured = organized in rows/columns (Excel, CSV). Unstructured = no fixed format (images, videos, emails).

Q3. Explain Mean, Median, Mode with solved examples. When to use each? (5 marks)

Answer: Mean = sum/count. Data: 10,20,30. Mean=60/3=20. Use for normal data. **Median** = middle when sorted. Odd: 5,8,12 → 8. Even: 2,4,6,8 → (4+6)/2=5. Use with outliers. **Mode** = most frequent. 3,5,3,7,3 → 3. Use for categorical data. Mean affected by outliers, median is not.

Quick Revision — Data Science

- Workflow: Collect → Clean → Explore → Model → Communicate
- Primary = collected by you | Secondary = already available
- Structured = tables | Unstructured = images/videos
- CSV = comma separated values, most common format
- Mean = average | Median = middle | Mode = most frequent
- Median better than mean with outliers
- NumPy = arrays | Pandas = DataFrames | Matplotlib = charts
- Always call plt.show() for charts

Unit 5

Computer Vision

Teaching machines to see and understand images — pixels, RGB, OpenCV, and CNN basics.

PART B — Subject Specific Skills 12 hrs theory + 18 hrs practical 4 Marks Theory

5.1 What is Computer Vision?

Simple Definition

Computer Vision (CV) is a field of AI that enables computers to "see" and understand images and videos, just like humans use their eyes.

Technical Definition

CV trains computers to interpret visual data (images/videos) by acquiring, processing, analysing, and understanding digital images to extract meaningful information.

Key Insight

You see a cat photo and instantly know it's a cat. A computer sees only a grid of numbers (pixel values like 128, 255, 0). CV teaches it what those numbers mean.

Applications

Application	Description	Example
Face Recognition	Detecting and identifying faces	Phone face unlock, Facebook tagging
Self-Driving Cars	Detecting roads, signs, pedestrians	Tesla Autopilot
Medical Imaging	Analysing X-rays, MRI, CT scans	COVID detection from chest X-ray
Agriculture	Crop disease from photos	Drones scanning fields
Augmented Reality	Overlaying digital info	Snapchat/Instagram filters
Document Processing	Reading text from images	Google Translate camera mode, OCR
E-Commerce	Visual product search	Google Lens — photo to search
Security	CCTV surveillance	Airport security facial matching

5.2 Image Basics — Pixels & Resolution

Pixel

A pixel (picture element) is the **smallest unit** of a digital image. Each pixel has a colour/brightness value. Millions of pixels together form the image.

Resolution

Number of pixels: Width × Height. Higher resolution = more detail. Examples: 720p (1280×720), 1080p (1920×1080), 4K (3840×2160).

Image Size: Width × Height × Channels bytes. A 1920×1080 RGB image = 1920×1080×3 ≈ 6.2 MB uncompressed.

5.3 Grayscale & RGB Images

Grayscale

1 channel, values 0 (black) to 255 (white). Shades of gray in between.

RGB (Colour Images)

3 channels — Red, Green, Blue. Each pixel has 3 values (0-255). Total colours: 256×256×256 = **16,777,216** (16.7 million).

Colour	R	G	B
Red	255	0	0
Green	0	255	0
Blue	0	0	255
White	255	255	255
Black	0	0	0
Yellow	255	255	0
Cyan	0	255	255
Magenta	255	0	255

Important

OpenCV reads images in **BGR** order (not RGB). Convert with `cv2.cvtColor(img, cv2.COLOR_BGR2RGB)`.

5.3b Activities: Exploring Computer Vision

Activity 1: Game — Emoji Scavenger Hunt (by Google)

What is it?

A web-based game where Google's AI shows you an emoji (e.g., 🍌, 🔑, 👟) and you must use your phone or computer camera to find that real-world object before the timer runs out.

How the CV Works

• **Real-time Object Detection:** The game uses a lightweight neural network (TensorFlow.js) running directly in the browser to analyze the live camera feed frame-by-frame.
• **Confidence Score:** The AI outputs probabilities (e.g., "Banana: 92%"). If the score passes a threshold, it matches the emoji and awards you points!

What to Do	What to Observe
Open `emojiscavengerhunt.withgoogle.com` on your mobile or webcam-enabled PC.	The camera feed opens with a timer. An emoji is shown at the top.
Point the camera at different objects around your room.	Watch the text at the bottom display what the AI "sees" in real time (e.g., "I think I see a chair, keyboard, laptop").

Activity 2: RGB Calculator

What is it?

An interactive color mixing tool (like `w3schools.com/colors/colors_rgb.asp`) where you slide Red, Green, and Blue values between 0 and 255 to create any color.

Try These Famous Mixes!

• **R=0, G=0, B=0** → **Black** (No light / all channels off)
• **R=255, G=255, B=255** → **White** (All light mixed / maximum value)
• **R=255, G=255, B=0** → **Yellow** (Red + Green)

• **R=0, G=255, B=255** → **Cyan** (Green + Blue)
• **R=255, G=0, B=255** → **Magenta** (Red + Blue)

Activity 3: Create Your Own Pixel Art

What is it?
Designing retro sprite art using tools like `piskelapp.com`. Drawing is done on a low-resolution grid (e.g., 16x16 or 32x32 pixels) rather than a high-resolution canvas.

Observation	Connection to Computer Vision
Zooming in very close	You see individual, solid square blocks of single colors (pixels).
Zooming out to normal size	The human brain merges the squares into a continuous shape (resolution and perception).
Changing a single pixel	Can change the entire "expression" of a character, illustrating why small pixel differences matter to CV models.

Activity 4: Convolutions & Image Kernels

What is a Convolution?
A **convolution** is a mathematical operation where a small grid of numbers (called a **kernel** or **filter**, usually 3x3) slides over an image, alters the pixel values, and creates a new, modified image. This is how computer vision models extract features like edges, blur, or sharpness.

Explore Interactive Kernels (setosa.io/ev/image-kernels)
Visit the website to see how sliding different 3x3 kernels over a picture instantly transforms it: • **Blur Filter:** Averages surrounding pixels, smoothing out high-frequency details.
• **Sharpen Filter:** Enhances the difference between neighboring pixels, making details pop.
• **Edge Detection (Sobel):** Highlights sudden changes in brightness, revealing borders of objects.

How Convolution Math Works (Exam Numerical Example)

Let's apply a 3x3 **Edge Detection Kernel** to a tiny 3x3 region of an image:

Image Pixels (Intensity)

10	10	10
10	80	80
10	80	80

3x3 Edge Kernel

-1	-1	-1
-1	8	-1
-1	-1	-1

Calculation of Output Pixel (for center element 80)
Multiply corresponding elements together and sum them up:

```
Output = (10 × -1) + (10 × -1) + (10 × -1) +
         (10 × -1) + (80 × 8)  + (80 × -1) +
         (10 × -1) + (80 × -1) + (80 × -1)

Output = (-10) + (-10) + (-10) + (-10) + (640) + (-80) + (-10) + (-80) + (-80)
```

`Output = -50 + 640 - 240 = 350` (normalized to 255 maximum → bright white pixel, indicating a strong edge!)

5.4 Computer Vision Tasks & Feature Extraction

Core Computer Vision Tasks

Computer Vision involves different tasks depending on what we want to achieve with an image. The main tasks are:

Task	Question Answered	Description	Example
Image Classification	"What is in the image?"	Labeling the entire image with a category.	Classifying an image as "Cat" or "Dog".
Object Detection	"What is where?"	Identifying multiple objects and drawing bounding boxes around them.	Detecting pedestrians and traffic lights in front of a self-driving car.
Image Segmentation	"Which pixels belong to what?"	Labeling every single pixel in the image to isolate exact shapes.	Highlighting the exact boundary of a tumor in an MRI scan.

Semantic vs Instance Segmentation

• **Semantic Segmentation:** Labels all objects of the same category with the same color (e.g., all cars are colored blue).
• **Instance Segmentation:** Distinguishes individual objects of the same category (e.g., Car 1 is blue, Car 2 is green, Car 3 is yellow).

Feature Extraction

Definition

Feature Extraction is the process of identifying and isolating key patterns or characteristics (like edges, corners, shapes, and textures) from an image, discarding irrelevant information (like background noise or lighting changes). It converts raw pixel data into compact, meaningful descriptors that AI models can use to recognize objects.

Types of Features

- **Low-Level Features:** Basic structures like **edges, lines, colors, and corners** (detected in early layers).
- **Mid-Level Features:** Simple combinations like **shapes, textures, and patterns** (e.g., circles, checkerboards).
- **High-Level Features:** Complex structures representing complete objects or parts (e.g., **eyes, nose, wheels, faces**).

How Features are Extracted

Method	How it Works	Example / Tool
Hand-Crafted Features (Traditional CV)	Human experts design mathematical rules to look for specific visual cues.	**Sobel Filter** (for edges), **Haar Cascades** (for faces in digital cameras).
Learned Features (Modern AI / Deep Learning)	The neural network automatically learns which features are important by looking at thousands of training images.	**Convolutional Neural Networks (CNNs)** automatically extract features using filters.

Real-Life Analogy

When you look at a face, your brain extracts features: the distance between the eyes, the shape of the jaw, and the width of the nose. You don't memorize every single skin cell. Similarly, Computer Vision models extract key coordinates and textures to identify a person.

Exam Tip

Feature extraction is crucial because a raw image contains millions of pixels, which is too much data for basic classifiers. Extracting features reduces the data size while keeping the most important information.

5.5 OpenCV

What is OpenCV?

Open-source library for computer vision and image processing. Install: `pip install opencv-python`. Import: `import cv2`.

PYTHON — READ, DISPLAY, SAVE IMAGE

```python
import cv2

img = cv2.imread("sample.jpg")          # Read image
print("Shape:", img.shape)               # (480, 640, 3) = H, W, C
cv2.imshow("Image", img)                 # Display in window
cv2.waitKey(0)                           # Wait for key press
cv2.destroyAllWindows()                 # Close all windows

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
print("Gray shape:", gray.shape)        # (480, 640) — only 2D

# Resize
resized = cv2.resize(img, (300, 200))  # (width, height)!

# Save image
cv2.imwrite("gray.jpg", gray)
```

img.shape returns (Height, Width, Channels). **cv2.resize()** takes (width, height) — don't mix up!

Drawing Shapes

PYTHON — DRAWING ON IMAGES

```python
cv2.line(img, (50,50), (450,50), (255,0,0), 3)      # Blue line
cv2.rectangle(img, (50,100), (300,300), (0,255,0), 3)  # Green rect
cv2.circle(img, (400,250), 80, (0,0,255), -1)     # Red filled circle (-1=filled)
cv2.putText(img, "AI", (100,400), cv2.FONT_HERSHEY_SIMPLEX, 2, (255,255,255), 3)
```

Key OpenCV Functions

Function	Purpose
`cv2.imread()`	Read image from file
`cv2.imshow()`	Display image in window
`cv2.waitKey(0)`	Wait for key press — **ESSENTIAL!**
`cv2.destroyAllWindows()`	Close all windows
`cv2.cvtColor()`	Convert colour space (BGR→Gray, BGR→RGB)
`cv2.resize()`	Resize image — takes (width, height)
`cv2.imwrite()`	Save image to file

`img.shape`	Get (height, width, channels)
`cv2.line/rectangle/circle/putText`	Draw shapes and text

OpenCV Pattern: "Read, Show, Wait, Destroy"

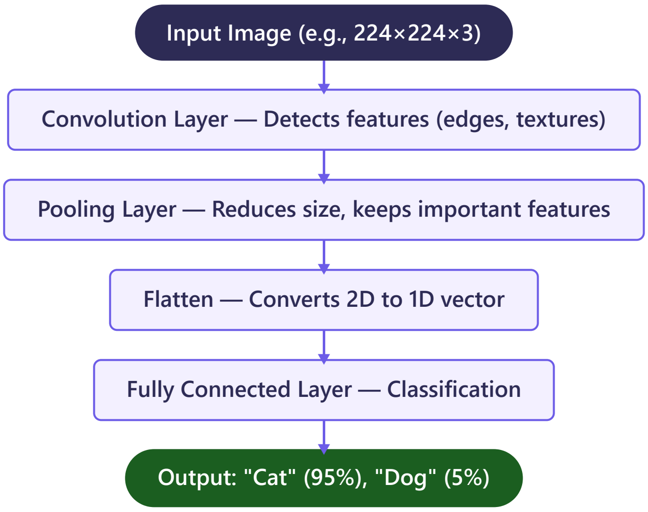
`imread()` → `imshow()` → `waitKey(0)` → `destroyAllWindows()`

5.6 CNN Basics (Optional/Extra Knowledge)

What is CNN?

A **Convolutional Neural Network** is a deep learning model for images. It automatically detects features (edges, shapes, objects) using kernels/filters that slide across the image.

CNN Architecture



Convolution = small filter (kernel) slides over image, multiplies values, produces feature map. Like a magnifying glass looking for specific patterns.

Uses: Image classification, face recognition, medical diagnosis, self-driving cars, handwriting recognition.

Activity: Testing a Convolutional Neural Network (CNN)

What is this Activity?

In this activity, you will test a pre-trained CNN model to see how it classifies real-world images. We explore two methods: a quick **No-Code Web Tool** and a **Python Code Script**.

Method 1: No-Code Web Tool (Google Teachable Machine)

Step	Action	Observation
1. Open Tool	Visit `teachablemachine.withgoogle.com` and select **Image Project**.	The interface shows two default classes (Class 1, Class 2).
2. Add Samples	Use your webcam or upload images to train Class 1 (e.g., "Pen") and Class 2 (e.g., "Notebook"). Capture 20-30 images for each from different angles.	The images are stored as training data.
3. Train Model	Click **Train Model**. Under the hood, a CNN model is configured and trained.	A progress bar runs as the CNN extracts features and learns to distinguish between classes.
4. Test & Evaluate	Show your pen or notebook to the camera in the **Preview** panel.	The CNN outputs prediction percentages in real-time (e.g., "Pen: 98%"). Watch it change as you move the object.

Method 2: Python Code (Test MobileNetV2 Pre-trained CNN)

Here is a complete Python program to classify any image using a pre-trained CNN model called **MobileNetV2** (which is trained on 1.2 million images to recognize 1,000 different object categories):

PYTHON – TEST PRE-TRAINED CNN

```python
1 from tensorflow.keras.applications.mobilenet_v2 import MobileNetV2, preprocess_input, decode_predictions
2 from tensorflow.keras.preprocessing import image
3 import numpy as np
4
5 # 1. Load the pre-trained CNN model
6 model = MobileNetV2(weights='imagenet')
7
8 # 2. Load and preprocess the test image (MobileNet requires 224x224 pixels)
9 img_path = 'cat.jpg'  # Replace with your image file path
10 img = image.load_img(img_path, target_size=(224, 224))
11 x = image.img_to_array(img)  # Convert image to numpy array
12 x = np.expand_dims(x, axis=0)  # Add batch dimension: shape becomes (1, 224, 224, 3)
13 x = preprocess_input(x)
14
15 # 3. Predict the category
16 preds = model.predict(x)
17
18 # 4. Print the top 3 predictions
19 print('Predicted:', decode_predictions(preds, top=3)[0])
```

Line 6: Loads the weights of MobileNetV2 trained on the ImageNet dataset. **Lines 8-13:** CNNs require fixed input dimensions. We resize the image to 224x224 pixels, convert it to a numerical array, and preprocess it. **Line 16:** Passes the image array through the CNN layers (Convolution, Pooling, FC) to compute class probabilities. **Line 19:** Decodes the probabilities to display the top 3 matched categories with their confidence scores.

5.7 Questions & Revision

Computer Vision | Pixel | Resolution | Grayscale | RGB | BGR | OpenCV | imread | imshow | waitKey | cvtColor | resize | imwrite | img.shape | CNN | Convolution | Kernel | Pooling | Object Detection | Segmentation

MCQs

1 Grayscale pixel values range:

a) 0 to 100

b) 0 to 255

c) -255 to 255

d) 1 to 256

Answer: b) 0 (black) to 255 (white).

2 RGB image has how many channels?

a) 1

b) 2

c) 3

d) 4

Answer: c) Red, Green, Blue = 3 channels.

3 OpenCV uses which colour format?

a) RGB

b) BGR

c) HSV

d) CMYK

Answer: b) BGR (Blue, Green, Red), not RGB.

4 RGB for White:

a) (0,0,0)

b) (255,0,0)

c) (255,255,255)

d) (128,128,128)

Answer: c) All channels at maximum = white.

5 Which converts colour to grayscale?

a) cv2.imread()

b) cv2.resize()

c) cv2.cvtColor()

d) cv2.imshow()

Answer: c) cv2.cvtColor(img, cv2.COLOR_BGR2GRAY).

6 Total RGB colours:

a) 256

b) 65,536

c) 16,777,216

d) 1,000,000

Answer: c) 256×256×256 = 16.7 million.

7 cv2.waitKey(0) does:

a) Closes windows

b) Waits 0 ms

c) Waits for any key

d) Saves image

Answer: c) Waits indefinitely for key press.

8 CNN stands for:

a) Computer Neural Network

b) Convolutional Neural Network

c) Connected Node Network

d) Calculated Number Network

Answer: b) Convolutional Neural Network.

Short & Long Answers

Q1. What is a pixel? How is resolution related? (2 marks)

Answer: Pixel = smallest unit of digital image with one colour value. Resolution = Width×Height in pixels. Higher resolution = more pixels = sharper image. 1920×1080 has ~2.1 million pixels.

Q2. Differentiate grayscale and RGB images. (2 marks)

Answer: Grayscale = 1 channel, 0-255 (black to white). RGB = 3 channels (Red, Green, Blue), each 0-255, represents 16.7 million colours. RGB uses 3x more memory.

Q3. Write a Python program to read an image, convert to grayscale, display both, and save. Explain each line. (5 marks)

Answer: `import cv2` (import library). `img = cv2.imread("photo.jpg")` (read colour image). `gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)` (convert BGR to grayscale). `cv2.imshow("Original", img)` (display colour). `cv2.imshow("Gray", gray)` (display grayscale). `cv2.waitKey(0)` (wait for key). `cv2.destroyAllWindows()` (close windows). `cv2.imwrite("gray.jpg", gray)` (save grayscale).

Quick Revision — Computer Vision

- CV = teaching computers to see and understand images
- Pixel = smallest unit | Resolution = W×H pixels
- Grayscale = 1 channel (0-255) | RGB = 3 channels (16.7M colours)
- OpenCV uses BGR, not RGB!
- Pattern: imread → imshow → waitKey → destroyAllWindows
- img.shape = (Height, Width, Channels)
- cv2.resize() takes (width, height) — opposite of shape!
- CNN: Convolution → Pooling → Flatten → FC → Output

Unit 6

Natural Language Processing

How machines understand, interpret, and generate human language — the bridge between human communication and AI.

PART B — Subject Specific Skills 25 hrs theory + 5 hrs practical 8 Marks

6.1 What is NLP?

Simple Definition

Natural Language Processing (NLP) is a branch of AI that helps computers understand, read, and make sense of human languages — just like how you read and understand this sentence.

Technical Definition

NLP is a sub-field of AI and computational linguistics that focuses on enabling machines to process, analyze, interpret, and generate natural (human) languages using algorithms and statistical models.

Why is NLP Important?

- **Language is the most natural form of communication** — we speak and write more naturally than we type code
- **Massive amounts of text data exist** — emails, social media, articles, reviews are all in natural language
- **Automation needs language understanding** — customer service, translation, search all require NLP
- **Accessibility** — NLP enables voice-controlled devices for people who cannot type

How Humans Communicate vs How Machines Process

Aspect	Human Communication	Machine Processing
Understanding	Intuitive, contextual, emotional	Rule-based or statistical pattern matching
Ambiguity	Handled naturally using context	Struggles without explicit rules/training
Learning	Learns from childhood through exposure	Requires large datasets and training
Speed	Relatively slower for large data	Can process millions of documents quickly
Emotions	Understands tone, sarcasm, humor	Finds it very difficult to detect

NLP Workflow

Text Input ("I love AI")

↓

Pre-processing (Tokenize, Clean)

↓

Feature Extraction (BoW, TF-IDF)

↓

AI Model (Classify / Predict)

↓

Output (Positive Sentiment)

Real-Life Examples of NLP

#	Example	How NLP is Used
1	**Google Search**	Understands your query and finds relevant pages, even with misspellings
2	**Google Translate**	Translates text between languages (Hindi to English)

3	**Alexa / Siri / Google Assistant**	Understands voice commands and responds with speech
4	**Gmail Spam Filter**	Reads email content and classifies as spam or genuine
5	**Auto-Correct on Phone**	Detects spelling errors and suggests corrections
6	**Netflix / YouTube**	Analyzes descriptions and reviews to recommend content
7	**Chatbots**	Understand questions and provide helpful answers
8	**Smart Compose (Gmail)**	Predicts and suggests next words as you type
9	**Grammarly**	Checks grammar, punctuation, and writing style
10	**Social Media Monitoring**	Companies analyze tweets to understand public opinion

---

Activity: Using Google Translate for Same-Spelling Words (Homonyms)

To understand how NLP models use context, try translating words with the same spelling but different meanings (homonyms) in Google Translate. Notice how the AI translates them differently based on the surrounding sentence.

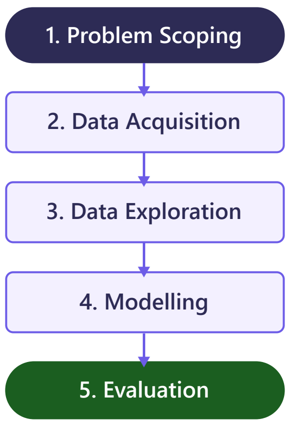
Try This in Google Translate!

Homonym	Sentence (English)	Google Translate Output (Hindi)	What it Shows
bank	"I deposited money in the bank."	"मैंने **बैंक** में पैसे जमा किये।"	Translated as financial institution.
	"The kids played near the river bank."	"बच्चे नदी के **किनारे** खेल रहे थे।"	Translated as shore/edge.
bat	"The bat flew out of the dark cave."	"**चमगादड़** अंधेरी गुफा से बाहर उड़ा।"	Translated as the nocturnal flying mammal.
	"He bought a brand new cricket bat."	"उसने एक नया क्रिकेट **बल्ला** खरीदा।"	Translated as the sports equipment.

Key Insight: If Google Translate translated word-by-word, it would fail. It uses an NLP transformer model to look at the **entire context** of the sentence to resolve ambiguity.

6.1b Revisiting AI Project Cycle for NLP

To build an NLP system, we follow the 5 stages of the **AI Project Cycle**. Let's map these stages to a project: **Building an Automatic Email Spam Classifier**.



1. **Problem Scoping:**
   - **Goal:** Detect and filter out annoying/dangerous spam emails.
   - **4 Ws Canvas:**
     - *Who is affected?* Email users.
     - *What is the problem?* Users waste time sorting spam, and face phishing risks.
     - *Where does it occur?* In the email inbox.
     - *Why solve it?* Improves productivity, enhances security.
2. **Data Acquisition:**
   - Collect thousands of email messages.

- Label each email manually as either `Spam` or `Ham` (genuine).
- Sources: Public datasets like the *Enron Spam Dataset* or user-reported spam logs.

3. **Data Exploration:**
   - Analyze the emails visually.
   - Notice patterns: e.g., Spam emails frequently contain uppercase words (`"FREE"` , `"URGENT"` , `"WIN"`) or specific symbols (`"$$$"`).
   - Check the dataset balance: Make sure you have a similar number of spam and ham emails.
4. **Modelling:**
   - **Pre-processing:** Perform text normalization (Tokenization, Stop Words removal, Lemmatization) to clean the email text.
   - **Feature Extraction:** Convert the clean text into numbers using Bag of Words or TF-IDF.
   - **Training:** Train a supervised classifier (e.g., Naive Bayes algorithm) on the numerical vectors.
5. **Evaluation:**
   - Test the classifier on new, unseen emails.
   - Calculate metrics like **Accuracy**, **Precision** (to avoid classifying genuine emails as spam), and **Recall**.
   - If performance is low, collect more data or fine-tune features.

6.2 Why Machines Struggle with Human Language

Human language is incredibly complex, flexible, and often illogical. Here are the specific challenges:

1. Ambiguity

> **Examples**
> **"I saw her duck."** — Meaning 1: I saw her pet duck (bird). Meaning 2: I saw her bend down.
> **"He went to the bank."** — Financial bank or river bank?
> **"The chicken is ready to eat."** — The chicken (food) is ready to be eaten, OR the chicken (animal) is hungry?

2. Sarcasm

> **Examples**
> **"Oh great, another Monday!"** — "great" sounds positive, but the person is annoyed.
> **"Thanks for the help, I really needed more problems."** — The person is actually upset.
> A machine might classify these as *positive* because of words like "great" and "thanks".

3. Idioms

> **Examples**
> **"It's raining cats and dogs"** = Heavy rain (not animals falling!)
> **"Break a leg"** = Good luck (not literally!)
> **"Piece of cake"** = Very easy
> **"Spill the beans"** = Reveal a secret
> **"Hit the nail on the head"** = Exactly right

4. Homophones & Homonyms

Type	Word 1	Word 2	Problem
Homophone	flour	flower	Sound same, different meaning — confuses speech systems
Homophone	write	right	Sound same: to write vs correct vs direction
Homophone	their	there/they're	Three words, same sound, different meanings
Homonym	bat (cricket)	bat (animal)	Same spelling, completely different meanings
Homonym	bark (tree)	bark (dog)	Same spelling, different meanings

5. Context, Slang & Multiple Languages

- **Context:** "Apple" — fruit or company? Depends on surrounding text
- **Slang:** "That's lit" = amazing, "GOAT" = Greatest Of All Time — not in dictionaries
- **Languages:** 7,000+ languages worldwide, each with unique grammar

Summary Table

Challenge	Example	Why Hard for Machines
Ambiguity	"I saw her duck"	Multiple meanings for same sentence
Sarcasm	"Oh great, another Monday!"	Literal words positive but meaning negative
Idioms	"Raining cats and dogs"	Cannot be understood word-by-word
Homophones	flour / flower	Sound same, different meaning
Homonyms	bat (cricket/animal)	Same spelling, different meanings
Context	"Apple" — fruit or company?	Needs surrounding text to determine
Slang	"That's lit" = amazing	Not in standard dictionaries

> **Exam Tip**
> Questions about why NLP is difficult are very common. Remember at least 4-5 challenges with examples.

6.3 Applications of NLP

Chatbots

> **Definition**
> A **chatbot** is a computer program that simulates conversation with human users using NLP.

Feature	Rule-Based Chatbot	AI-Based Chatbot
How it works	Follows pre-written if-else rules	Uses ML and NLP to understand intent
Flexibility	Can only answer pre-defined questions	Handles new and unexpected questions
Learning	Cannot learn from conversations	Improves over time with more data
Example	Basic FAQ bot on a website	Google Assistant, ChatGPT, Alexa

---

Activity: Designing a Rule-Based vs. AI Chatbot

> **Activity Goal**
> To understand how chatbots work under the hood by mapping the logic of a simple **Rule-Based Chatbot** and testing where it fails compared to an **AI Chatbot**.

Task 1: Map the Logic Tree of a School Inquiry Bot

User: "Hello"

↓

Bot: "Welcome! Choose 1 (Admissions), 2 (Fees), or 3 (Timings)"

Option 1
User: `"1"`
Bot: "Admissions are open! Apply at school.com"

Option 2
User: `"2"`
Bot: "Fees can be paid online at portal.com"

Option 3
User: `"3"`
Bot: "School hours are 8:00 AM to 2:00 PM"

Task 2: Test the Fragility (Where it fails)

Imagine a user types: **"Can you tell me when school finishes?"**

- 🔴 **Rule-Based Chatbot:** Since the input is not exactly `"3"`, it falls into the else condition: *"Sorry, I did not understand that. Please enter 1, 2, or 3."* (Fails because it has no language understanding).
- 🟢 **AI Chatbot:** Processes the sentence using NLP, extracts the intent (`"school timings"`), matches it to "Timings", and replies: *"School is open from 8:00 AM to 2:00 PM."*

DIY: Write a Simple Rule-Based Chatbot in Python

PYTHON – RULE-BASED CHATBOT

```python
def school_bot():
    print("Bot: Hello! Welcome to School Helpdesk.")
    while True:
        user_input = input("You (Type 1 for Admissions, 2 for Fees, 3 for Timings, or 'exit'): ").strip().lower()

        if user_input == 'exit':
            print("Bot: Thank you for visiting! Goodbye.")
            break
        elif user_input == '1':
            print("Bot: Admissions are open for Session 2026-27.")
        elif user_input == '2':
            print("Bot: Please pay fees online on our official portal.")
        elif user_input == '3':
            print("Bot: School hours are 8:00 AM to 2:00 PM.")
        else:
            print("Bot: Sorry, I didn't get that. Please enter 1, 2, 3, or 'exit'.")

# Run the bot
school_bot()
```

> Run this code in Python. Try entering inputs other than 1, 2, or 3 to observe how rule-based chatbots fail to handle conversational language.

Sentiment Analysis

> **Definition**
> **Sentiment Analysis** uses NLP to identify the emotional tone of text as **Positive**, **Negative**, or **Neutral**.

Sentiment	Indicator Words	Example
Positive	love, great, amazing, excellent, best	"This phone has an amazing camera!"

Negative	hate, terrible, worst, awful, boring	"Terrible service, will never come back."
Neutral	Factual statements without opinion	"The meeting is at 3 PM."

Uses: Product reviews (Amazon), social media monitoring, customer feedback, stock market prediction, movie ratings.

Other Applications

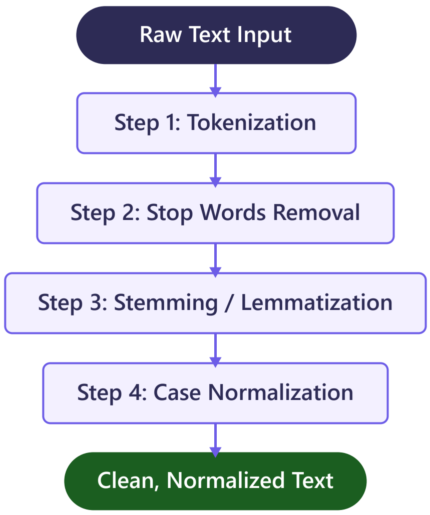
- **Google Translate:** Translates between languages using Neural Machine Translation
- **Voice Assistants:** Siri, Alexa use Speech-to-Text and Text-to-Speech
- **Spam Detection:** Gmail reads emails for spam patterns ("free", "winner", "click here")
- **Auto-Correct/Complete:** Phone changes "teh" to "the"; Google predicts your search
- **Text Summarization:** NLP reads long articles and generates short summaries
- **Smart Compose:** Gmail suggests how to complete your sentences

---

6.4 Text Normalization (Pre-processing)

> **Definition**
> **Text Normalization** is cleaning and standardizing raw text so a machine can process it. Involves removing noise like capitals, punctuation, and common words.

Five Steps of Text Normalization



Step 1: Tokenization

> **Definition**
> **Tokenization** splits text into smaller units called **tokens** (words or sentences).

> **Example**
> **Input:** "AI is amazing and powerful"
> **Output:** `["AI", "is", "amazing", "and", "powerful"]` — 5 tokens

Step 2: Stop Words Removal

> **Definition**
> **Stop words** are common words that carry little meaning: is, am, are, the, a, an, in, on, at, of, for, to, and, but, or.

> **Example**
> **Input:** `["AI", "is", "amazing", "and", "powerful"]`
> **After removal:** `["AI", "amazing", "powerful"]`

Step 3: Stemming

> **Definition**

Stemming reduces words to root form by removing suffixes. Fast but may give **non-words**.

Original	After Stemming	Valid?
running	runn	No
playing	play	Yes
flies	fli	No
happiness	happi	No

Step 4: Lemmatization

Definition

Lemmatization reduces words to **dictionary form** (lemma). Always produces a **valid word**.

Original	After Lemmatization	Valid?
running	run	Yes
flies	fly	Yes
better	good	Yes
mice	mouse	Yes

Stemming vs Lemmatization

Feature	Stemming	Lemmatization
Method	Chops off suffixes	Uses dictionary and grammar
Result	May not be a real word	Always a valid word
Speed	Faster	Slower
Accuracy	Less accurate	More accurate
"running"	runn	run
"better"	better	good

Complete Worked Example — All 5 Steps

Solved Example

Original: "The students ARE Running and Playing in the GARDEN happily."

Step 1 — Tokenization: ["The", "students", "ARE", "Running", "and", "Playing", "in", "the", "GARDEN", "happily"]
Step 2 — Lowercase: ["the", "students", "are", "running", "and", "playing", "in", "the", "garden", "happily"]
Step 3 — Stop Words: ["students", "running", "playing", "garden", "happily"]
Step 4 — Lemmatization: ["student", "run", "play", "garden", "happily"]
Clean text ready for NLP analysis!

Memory Trick — "Teachers Should Study Lessons Carefully"

Tokenization → **S**top words → **S**temming/Lemmatization → **L**owercase → **C**lean text

6.5 Bag of Words (BoW)

Definition

Bag of Words converts text into numerical vectors by counting word frequency. Ignores word order — treats text like a "bag" of words.

Complete Worked Example

Doc 1: "I love AI" | Doc 2: "AI is great" | Doc 3: "I love great AI"

Vocabulary: {I, love, AI, is, great}

Document	I	love	AI	is	great
Doc 1	1	1	1	0	0
Doc 2	0	0	1	1	1
Doc 3	1	1	1	0	1

Limitations: (1) Ignores word order — "Dog bites man" = "Man bites dog". (2) No semantic meaning. (3) Sparse vectors. (4) Common words dominate.

Key Exam Point

BoW's biggest limitation: **ignores word order**. "Dog bites man" and "Man bites dog" have identical vectors despite opposite meanings.

6.6 TF-IDF

Formulas

TF(t,d) = (Times term t in doc d) / (Total terms in doc d)
IDF(t) = log(Total docs / Docs containing t)
TF-IDF = TF × IDF

Words appearing in every document: IDF = log(N/N) = 0, so TF-IDF = 0. Common words automatically get zero score!

Feature	Bag of Words	TF-IDF
Measures	Word count (frequency)	Word importance (frequency + rarity)
Common words	High values (problem!)	Low values (solved!)
Rare words	Same as common	Higher importance

Memory Trick for TF-IDF

TF = "How much does THIS document talk about this word?" (local)
IDF = "How RARE is this word across ALL documents?" (global)
Important if appears a LOT here but RARELY elsewhere.

6.7 NLTK (Optional/Practical)

Definition

NLTK is a free Python library for NLP: tokenization, stemming, lemmatization, stop word removal.

PYTHON — TOKENIZATION

```python
from nltk.tokenize import word_tokenize
text = "AI is amazing and powerful"
tokens = word_tokenize(text)
print(tokens)  # ['AI', 'is', 'amazing', 'and', 'powerful']
```

PYTHON — STOP WORDS + STEMMING

```python
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer
```

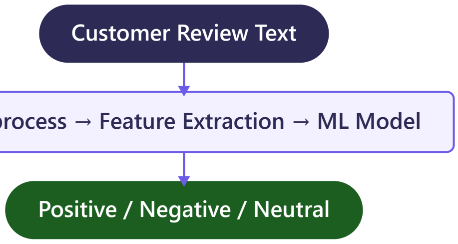
```python
stop_words = set(stopwords.words('english'))
filtered = [w for w in tokens if w.lower() not in stop_words]
print(filtered)  # ['AI', 'amazing', 'powerful']

stemmer = PorterStemmer()
for w in ["running", "played", "flies"]:
    print(w, "→", stemmer.stem(w))  # runn, play, fli
```

6.8 Sentiment Analysis

Definition

Sentiment Analysis (Opinion Mining) uses NLP to classify text as Positive, Negative, or Neutral.



Uses: Amazon reviews, Twitter trends, customer feedback, stock prediction, movie ratings.

6.9 Questions & Revision

NLP · Tokenization · Stop Words · Stemming · Lemmatization · Bag of Words · TF-IDF · Sentiment Analysis · Chatbot · Ambiguity · Sarcasm · Idioms · Homophones · Homonyms · NLTK · Machine Translation · Voice Assistant · Pre-processing

MCQs

1 Breaking text into words is called:

a) Stemming

b) Lemmatization

c) Tokenization

d) Normalization

Answer: c) Tokenization splits text into tokens.

2 Which always gives a valid dictionary word?

a) Stemming

b) Lemmatization

c) Tokenization

d) BoW

Answer: b) Lemmatization always produces valid words.

3 Stemming of "running" produces:

a) run

b) runn

c) running

d) ran

Answer: b) Stemming gives "runn" (not valid). Lemmatization gives "run".

4 BoW's major limitation:

a) Cannot count

b) Ignores word order

c) Can't process English

d) Requires images

Answer: b) BoW ignores word order.

5 In TF-IDF, a word in every document gets:

a) High score

b) Score of zero

c) Negative score

d) Score of one

Answer: b) IDF = log(N/N) = 0. TF-IDF = 0.

6 "Oh great, another Monday!" is hard because of:

a) Ambiguity

b) Sarcasm

c) Idioms

d) Homophones

Answer: b) Sarcasm — "great" sounds positive but person is annoyed.

7 Sentiment Analysis classifies text into:

a) True/False

b) Positive/Negative/Neutral

c) Subject/Object

d) Noun/Verb

Answer: b) Positive, Negative, or Neutral.

8 Which is NOT an NLP challenge?

a) Ambiguity

b) Sarcasm

c) Math calculations

d) Idioms

Answer: c) Math is not a language challenge.

Short Answer Questions

Q1. Define NLP. Give two applications. (2 marks)

Answer: NLP is a branch of AI that enables computers to understand human language. Applications: (1) Google Translate converts Hindi to English. (2) Chatbots on websites answer customer questions automatically.

Q2. Differentiate Stemming and Lemmatization. (3 marks)

Answer: Stemming chops suffixes, may give non-words ("running"→"runn"). Lemmatization uses dictionary, always gives valid words ("running"→"run"). Lemmatization is more accurate but slower.

Q3. What is Bag of Words? State its limitation. (2 marks)

Answer: BoW converts text to numerical vectors by counting word frequency. Major limitation: ignores word order — "Dog bites man" and "Man bites dog" produce identical vectors.

Long Answer Questions

Q1. Explain Text Normalization steps with example. (5 marks)

Answer: (1) **Tokenization:** Split into words. "AI is great" → ["AI","is","great"]. (2) **Case Normalization:** Lowercase. (3) **Stop Words:** Remove common words like "is". (4) **Stemming:** Reduce to root (may give non-words). (5) **Lemmatization:** Reduce to dictionary form (always valid). Example: "The Students ARE Playing" → ["student","play"].

Q2. Why do machines struggle with human language? Give 5 challenges. (5 marks)

Answer: (1) **Ambiguity:** "I saw her duck" has 2 meanings. (2) **Sarcasm:** "Oh great, another Monday!" means opposite. (3) **Idioms:** "Raining cats and dogs" = heavy rain. (4) **Homophones:** flour/flower confuse speech systems. (5) **Context:** "Apple" could be fruit or company.

Quick Revision — NLP

- NLP = AI for understanding human language
- Challenges: Ambiguity, Sarcasm, Idioms, Homophones, Homonyms, Context, Slang
- Steps: Tokenization → Stop Words → Stemming/Lemmatization → Lowercase
- Stemming = crude, non-words | Lemmatization = dictionary, valid words
- BoW = count frequency, ignores order | TF-IDF = TF×IDF, measures importance
- Sentiment Analysis: Positive, Negative, Neutral

Unit 7

Evaluation

Measuring AI model performance — Confusion Matrix, Accuracy, Precision, Recall, F1 Score with solved examples.

PART B — Subject Specific Skills 15 Hours 8 Marks

7.1 What is Model Evaluation?

Simple Definition

Model Evaluation is checking how well an AI model's predictions match reality — like checking exam answers against the answer key.

Why it matters: (1) Trust before deploying in real life. (2) Know where model goes wrong. (3) Compare models. (4) Wrong predictions can be dangerous.

7.2 Underfitting, Good Fit, Overfitting

Feature	Underfitting	Good Fit	Overfitting
Training performance	Poor	Good	Excellent
Test performance	Poor	Good	Poor
Model complexity	Too simple	Just right	Too complex
Student analogy	Didn't study at all	Understood concepts	Memorized without understanding
Fix	More complex model, more features	No fix needed!	Simpler model, more data

Key Point

Underfitting = poor on BOTH. **Overfitting** = great training, POOR test. **Good Fit** = good on BOTH.

Prevent Overfitting: More data, simpler model, cross-validation, early stopping.

7.3 TP, TN, FP, FN

	Actually Positive	Actually Negative
Predicted Positive	**True Positive (TP)**	**False Positive (FP) — Type I Error**
Predicted Negative	**False Negative (FN) — Type II Error**	**True Negative (TN)**

Term	Prediction	Actual	Correct?	Disease Example
TP	Sick	Sick	YES	Correctly detected disease
TN	Healthy	Healthy	YES	Correctly cleared patient
FP	Sick	Healthy	NO	False alarm — unnecessary worry
FN	Healthy	Sick	NO	Missed case — DANGEROUS!

Multiple Scenarios

Scenario	TP	FP (False Alarm)	FN (Missed)
Spam Filter	Spam caught correctly	Good email marked spam	Spam reaches inbox
Fire Alarm	Alarm rings, fire exists	Alarm rings, no fire	No alarm but fire exists!
COVID Test	Positive, has COVID	Positive, no COVID	Negative, has COVID!

Memory Trick

First word = RIGHT or WRONG: **True** = Correct, **False** = Wrong.
Second word = what model SAID: **Positive** = YES, **Negative** = NO.

7.4 Confusion Matrix

Definition

A **Confusion Matrix** is a 2×2 table of TP, TN, FP, FN. Diagonal = correct. Off-diagonal = errors.

Solved Example: Spam Filter

Building the Matrix

100 emails: 50 spam, 50 legit. Of 50 spam: 45 caught, 5 missed. Of 50 legit: 42 correct, 8 wrongly flagged.
TP=45, FN=5, FP=8, TN=42. Total: 100.

	Predicted Spam	Predicted Not Spam
Actual Spam	**TP = 45**	**FN = 5**
Actual Not Spam	**FP = 8**	**TN = 42**

Solved Example: Disease Detection

Building the Matrix

200 patients: 80 diseased, 120 healthy. Model predicted 90 as diseased: 70 correct, 20 wrong.
TP=70, FP=20, FN=80-70=10, TN=120-20=100. Total: 200.

	Predicted Diseased	Predicted Healthy
Actual Diseased	**TP = 70**	**FN = 10**
Actual Healthy	**FP = 20**	**TN = 100**

Activity: Containment Zone Prediction Model

The Scenario: To control an epidemic outbreak, a district administration deploys an AI model that predicts whether a locality should be declared a **"Containment Zone"** (Positive Class) or a **"Safe Zone"** (Negative Class). Here is the actual data and the model predictions for 10 local areas:

Locality	Actual Status	Model Prediction	Classification
L1	Containment	Containment	**True Positive (TP)** — Correctly predicted high-risk
L2	Safe	Safe	**True Negative (TN)** — Correctly predicted safe
L3	Containment	Safe	**False Negative (FN)** — Missed hazard! (Dangerous)
L4	Safe	Containment	**False Positive (FP)** — False alarm (Lockdown panic)
L5	Containment	Containment	**True Positive (TP)** — Correctly predicted high-risk
L6	Safe	Safe	**True Negative (TN)** — Correctly predicted safe
L7	Containment	Containment	**True Positive (TP)** — Correctly predicted high-risk
L8	Safe	Safe	**True Negative (TN)** — Correctly predicted safe
L9	Safe	Containment	**False Positive (FP)** — False alarm
L10	Containment	Safe	**False Negative (FN)** — Missed hazard!

Step-by-Step Matrix Count

- **True Positives (TP):** Actual Containment & Predicted Containment = **3** (L1, L5, L7)
- **True Negatives (TN):** Actual Safe & Predicted Safe = **3** (L2, L6, L8)

- **False Positives (FP):** Actual Safe & Predicted Containment = **2** (L4, L9)
- **False Negatives (FN):** Actual Containment & Predicted Safe = **2** (L3, L10)

Resulting Confusion Matrix

	Predicted Containment Zone	Predicted Safe Zone
Actual Containment Zone	**TP = 3**	**FN = 2**
Actual Safe Zone	**FP = 2**	**TN = 3**

7.5 Evaluation Metrics with Formulas

All Formulas

Accuracy = (TP + TN) / (TP + TN + FP + FN) — "Of ALL predictions, how many correct?"
Precision = TP / (TP + FP) — "Of predicted positives, how many actually positive?"
Recall = TP / (TP + FN) — "Of actual positives, how many found?"
F1 Score = 2 × (P × R) / (P + R) — Harmonic mean of Precision and Recall.

Worked Example (Spam Filter: TP=45, TN=42, FP=8, FN=5)

- **Accuracy** = (45+42)/100 = 87/100 = **87%**
- **Precision** = 45/(45+8) = 45/53 = **84.9%**
- **Recall** = 45/(45+5) = 45/50 = **90%**
- **F1** = 2×(0.849×0.90)/(0.849+0.90) = **87.4%**

When is Which Metric Important?

Scenario	Metric	Reason
Medical Diagnosis	**Recall**	Missing sick patient (FN) is life-threatening
Spam Filter	**Precision**	Good email as spam (FP) loses important emails
Fraud Detection	**Recall**	Missing fraud (FN) causes financial loss
Airport Security	**Recall**	Missing weapon (FN) is catastrophic

Formula Memory: "A-P-R-F"

Accuracy = **A**ll correct / **A**ll total
Precision = **P**redicted positives right / **P**redicted positives
Recall = **R**eal positives found / **R**eal positives total
F1 = **F**usion of P and R

When Accuracy is Misleading

The Accuracy Trap

95 healthy, 5 sick. Model always says "Healthy": Accuracy = 95/100 = 95%! But TP=0, missed ALL sick patients. Need Precision/Recall/F1 for imbalanced data.

Complete Solved Problem

Fire Alarm: TP=25, FN=5, FP=15, TN=105 (Total=150)

Accuracy = (25+105)/150 = 130/150 = **86.7%**
Precision = 25/(25+15) = 25/40 = **62.5%**
Recall = 25/(25+5) = 25/30 = **83.3%**
F1 = 2×(0.625×0.833)/(0.625+0.833) = **71.4%**
Analysis: Recall should be higher for fire alarms — missing fire is fatal.

Numerical: TP=60, FP=15, FN=10, TN=115 (Total=200)

> Accuracy = 175/200 = **87.5%**
> Precision = 60/75 = **80%**
> Recall = 60/70 = **85.7%**
> F1 = 2×(0.80×0.857)/(0.80+0.857) = **82.8%**

Activity: Practice Evaluation

Test your understanding of model evaluation metrics by solving these practice scenarios. Work out the calculations manually, then verify your answers using the step-by-step solutions below.

Practice Problem 1: Forest Fire Detection Model

An AI satellite monitoring system is evaluated over a dry season. Out of 200 observations, it predicts forest fires (Positive Class) vs. normal green cover (Negative Class).

Given Data: `TP = 45`, `FN = 5`, `FP = 15`, `TN = 135`.

1. Calculate Accuracy, Precision, Recall, and F1 Score.
2. Identify which metric is most critical for this scenario and explain why.

Solution 1 — Step-by-Step

- **Total Observations:** $TP + TN + FP + FN = 45 + 135 + 15 + 5 = 200$
- **Accuracy:** $\frac{TP + TN}{\text{Total}} = \frac{45 + 135}{200} = \frac{180}{200} = \mathbf{90\%}$
- **Precision:** $\frac{TP}{TP + FP} = \frac{45}{45 + 15} = \frac{45}{60} = \mathbf{75\%}$
- **Recall:** $\frac{TP}{TP + FN} = \frac{45}{45 + 5} = \frac{45}{50} = \mathbf{90\%}$
- **F1 Score:** $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = 2 \times \frac{0.75 \times 0.90}{0.75 + 0.90} = 2 \times \frac{0.675}{1.65} \approx \mathbf{81.8\%}$
- **Most Critical Metric: Recall** is critical because a False Negative ($FN = 5$) means 5 forest fires went completely undetected, which could lead to massive loss of life and resources. A False Positive ($FP = 15$) only results in a false alarm verification check.

Practice Problem 2: Fraud Detection (Imbalanced Data)

A bank tests a new machine learning model to flag credit card fraud (Positive Class) vs. normal transactions (Negative Class) out of a batch of 1,000 transactions.

Given Data: `TP = 18`, `FN = 2`, `FP = 80`, `TN = 900`.

1. Calculate the Accuracy of the model.
2. Calculate the Precision, Recall, and F1 Score.
3. Explain why the Accuracy score is misleading here.

Solution 2 — Step-by-Step

- **Total Transactions:** $18 + 900 + 80 + 2 = 1000$
- **Accuracy:** $\frac{18 + 900}{1000} = \frac{918}{1000} = \mathbf{91.8\%}$
- **Precision:** $\frac{18}{18 + 80} = \frac{18}{98} \approx \mathbf{18.4\%}$
- **Recall:** $\frac{18}{18 + 2} = \frac{18}{20} = \mathbf{90\%}$
- **F1 Score:** $2 \times \frac{0.184 \times 0.90}{0.184 + 0.90} = 2 \times \frac{0.1656}{1.084} \approx \mathbf{30.6\%}$
- **Why Accuracy is Misleading:** The Accuracy of 91.8% looks excellent, but the model has a very high number of False Positives ($FP = 80$) compared to correct frauds caught ($TP = 18$), leading to a terrible Precision of 18.4%. The F1 Score (30.6%) gives a much more realistic picture of the model's poor balanced performance on this imbalanced dataset.

7.6 Questions & Revision

Confusion Matrix · TP · TN · FP · FN · Accuracy · Precision · Recall · F1 Score · Overfitting · Underfitting · Type I Error · Type II Error · Harmonic Mean

MCQs

1 Great on training, poor on test =

a) Underfitting

b) Good Fit

c) Overfitting

d) Perfect

Answer: c) Overfitting.

2 Precision formula:

a) TP/(TP+FN)

b) TP/(TP+FP)

c) (TP+TN)/Total

d) 2PR/(P+R)

Answer: b) TP/(TP+FP).

3 Medical diagnosis needs:

a) Precision

b) Accuracy

c) Recall

d) F1

Answer: c) Recall — missing sick patient is fatal.

4 TP=30, TN=50, FP=10, FN=10. Accuracy:

a) 70%

b) 75%

c) 80%

d) 85%

Answer: c) (30+50)/100 = 80%.

5 FN in disease detection:

a) Healthy correctly ID'd

b) Sick correctly ID'd

c) Healthy wrongly sick

d) Sick wrongly healthy

Answer: d) FN = missed disease.

6 F1 is ____ mean:

a) Arithmetic

b) Geometric

c) Harmonic

d) Weighted

Answer: c) Harmonic mean.

7 Spam filter needs:

a) Recall

b) Precision

c) Only Accuracy

d) None

Answer: b) Precision — don't lose good emails.

8 TP means:

a) Predicted positive, wrong

b) Predicted negative, correct

c) Predicted positive, correct

d) Predicted negative, wrong

Answer: c) TP = predicted positive AND actually positive.

9 TP=50, TN=40, FP=5, FN=5. Accuracy:

a) 85%

b) 90%

c) 95%

d) 80%

Answer: b) 90/100 = 90%.

10 Recall formula:

a) TP/(TP+FP)

b) TP/(TP+FN)

c) TN/(TN+FN)

d) FP/(FP+TN)

Answer: b) TP/(TP+FN).

Short Answer Questions

Q1. Define Overfitting and Underfitting. (2 marks)

Answer: Overfitting = model memorizes training data, great on training but poor on test (like memorizing answers). **Underfitting** = model too simple, poor on both (like not studying).

Q2. What is a Confusion Matrix? (2 marks)

Answer: A 2×2 table showing TP, TN, FP, FN. Diagonal (TP,TN) = correct. Off-diagonal (FP,FN) = errors.

Q3. Differentiate Precision and Recall. (2 marks)

Answer: Precision = TP/(TP+FP), quality of positive predictions. Recall = TP/(TP+FN), coverage of actual positives. Precision for spam filter, Recall for medical diagnosis.

Long Answer Questions

Q1. Explain all metrics with formulas. Why is accuracy misleading? (5 marks)

Answer: Accuracy=(TP+TN)/Total. Precision=TP/(TP+FP). Recall=TP/(TP+FN). F1=2PR/(P+R). Accuracy misleading with imbalanced data: 95 healthy + 5 sick, always predicting "healthy" = 95% accuracy but misses ALL sick patients.

Q2. Given TP=40, FP=10, FN=5, TN=45. Calculate all metrics. (5 marks)

Answer: Accuracy=(40+45)/100=**85%**. Precision=40/50=**80%**. Recall=40/45=**88.9%**. F1=2(0.80×0.889)/(0.80+0.889)=**84.2%**.

Quick Revision — Evaluation

- Underfitting = poor both | Overfitting = great train, poor test | Good Fit = good both
- TP = correct positive | TN = correct negative | FP = false alarm | FN = missed
- Accuracy = (TP+TN)/Total | Precision = TP/(TP+FP) | Recall = TP/(TP+FN)
- F1 = 2PR/(P+R) = harmonic mean
- Precision when FP costly | Recall when FN costly | Accuracy misleading with imbalanced data

C

Practical Work & Exam Guide

Complete practical exam guide — 18 programs with code, output, line explanations, viva questions, and scoring strategies.

PART C   Practical File: 15 marks   Practical Exam: 15 marks   Viva: 5 marks

Official CBSE Curriculum Resources

Access the official publications, study materials, and practical notebooks recommended by the CBSE board for the Class 10 Artificial Intelligence curriculum:

📘 CBSE Facilitator Handbook (Class X) — What it means:

This is the **official textbook and curriculum guide** published directly by the CBSE board for Subject Code 417. It contains official board-approved explanations for all theory units (AI Domains, Smart Cities, CV, NLP) along with suggested classroom discussion sheets, teacher lesson plans, and the official assessment rubrics for school board exams.

Open Handbook PDF →

🐍 AI Activities & Jupyter Notebooks — What it means:

This is the **official digital laboratory repository** containing interactive Python files (`.ipynb`) for the practical modules. It includes pre-written code for the 8 suggested practical programs (NumPy, Matplotlib, Pandas, OpenCV) along with dummy databases (`data.csv`) and test images (`sample.jpg`) so students can execute, edit, and experiment with the code instantly on their browser or local system without setup errors.

Access Jupyter Notebooks →

C.1 Practical Exam Structure

Component	Details	Marks
Practical File	Minimum 15 programs with aim, code, output, teacher signature	15
Practical Exam — Python	Write & execute a Python program	5
Practical Exam — Data Science	NumPy/Pandas/Matplotlib program	5
Practical Exam — Computer Vision	OpenCV program	5
Viva Voce	Oral questions on theory & code	5
Total Practical		**35**

Tips to Score Full Marks

1. Practice all programs from memory
2. Understand each line — prepare for "what does this line do?"
3. Keep practical file complete, neat, with teacher signatures
4. Keep sample CSV files and images in your working directory
5. Always write import statements first
6. Never forget `plt.show()` for charts and `cv2.waitKey(0)` for images

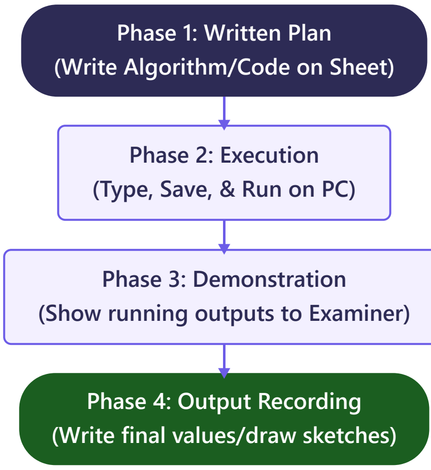
7. Be ready for modified questions (same concept, different data)

> **Common Mistakes to Avoid**
> (1) Incomplete practical file. (2) Forgetting imports. (3) Missing plt.show(). (4) Missing cv2.waitKey(0). (5) CSV/image file not in working directory. (6) Panicking during viva.

Comprehensive Practical Exam Preparation Guide

1. Step-by-Step Practical Exam Workflow

On exam day, you will be allocated a computer terminal and given a question sheet containing 3 practical coding tasks (Python, Data Science, CV). You must follow this 4-phase workflow:



2. Domain-Wise Coding Expectations

Domain / Unit	What You Must Implement	What the Examiner Verifies
Unit 3: Advance Python	Logic building using loops, conditionals, list operations, or functions.	Code execution without syntax/indentation errors and correct math logic.
Unit 4: Data Science	Creating a NumPy array, calculating stats (mean, median, mode), or generating line/scatter charts.	Proper axes labeling, plot title, chart markers, and final display via `plt.show()`.
Unit 5: Computer Vision	Reading a local image, checking properties (shape/dimensions), converting colors, or drawing shapes.	Correct image reading path, display window loading, and closing via `cv2.waitKey(0)`.

3. Strategies and Suggestions to Score 35/35

- 💡 **Prepare Assets Early:** Make sure a test dataset file (e.g., `data.csv`) and a test image (e.g., `sample.jpg`) are saved directly in your execution directory before writing code.
- 💡 **Resolve Package Errors:** If a package is missing on your terminal, open the terminal/command prompt and quickly install it:

```
> pip install numpy pandas matplotlib opencv-python
```

- 💡 **Write Clean Code:** Add short, 1-line comments (using `#`) to explain your logic. Examiners love documented code and it helps you stand out during evaluation.
- 💡 **Double-Check Terminators:** Always add `plt.show()` for plots, and `cv2.waitKey(0)` followed by `cv2.destroyAllWindows()` for OpenCV image windows to avoid freeze issues.

4. Sample Practical Exam Questions with Ideas

> **Sample Question 1 (Unit 3: Python Core)**
> **Task:** Write a Python program to input 5 numbers into a list and display their product.
> **Practical Idea:**
> 1. Initialize an empty list: `num_list = []`.
> 2. Use a `for` loop to take 5 float inputs using `input()`.
> 3. Loop through the list, multiplying the numbers together to find the final product.

Expected Output: `Product of elements is: 120.0`

Sample Question 2 (Unit 4: Data Science)

Task: Write a program using Matplotlib to display a bar chart comparing the AI scores of three students: Amit (85), Ritu (92), and Sameer (78). Label the axes and set a title.

Practical Idea:

1. Import matplotlib: `import matplotlib.pyplot as plt`.
2. Set names list `['Amit', 'Ritu', 'Sameer']` and marks list `[85, 92, 78]`.
3. Use `plt.bar(names, marks, color='teal')`.
4. Add labels: `plt.xlabel('Students')`, `plt.ylabel('AI Marks')`, and `plt.show()`.

Sample Question 3 (Unit 5: Computer Vision)

Task: Write a program using OpenCV to read an image file named `photo.jpg`, display its width and height, convert it to grayscale, and save the grayscale image as `gray_photo.jpg`.

Practical Idea:

1. Import opencv: `import cv2`.
2. Load image: `img = cv2.imread('photo.jpg')`.
3. Print dimensions: `h, w, c = img.shape`, then print `w` and `h`.
4. Convert to grayscale: `gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)`.
5. Save image: `cv2.imwrite('gray_photo.jpg', gray)`.

C.2 Complete Solved Programs (CBSE Suggested)

Program 1: Add Elements of Two Lists

PYTHON

```python
1 list1 = [10, 20, 30, 40, 50]
2 list2 = [5, 15, 25, 35, 45]
3 result = [a + b for a, b in zip(list1, list2)]
4 print("Sum:", result)  # [15, 35, 55, 75, 95]
```

Line 3: `zip()` pairs elements. List comprehension adds each pair. Result: [15, 35, 55, 75, 95].

Program 2: Mean, Median, Mode

PYTHON

```python
import numpy as np
from scipy import stats
data = [10, 20, 30, 20, 40, 50, 20]
print("Mean:", np.mean(data))      # 27.14
print("Median:", np.median(data))  # 20.0
print("Mode:", stats.mode(data).mode)  # 20
```

Program 3: Line Chart (2,5) to (9,10)

PYTHON

```
import matplotlib.pyplot as plt
plt.plot([2,9], [5,10], marker="o", color="blue")
plt.xlabel("X"); plt.ylabel("Y"); plt.title("Line Chart"); plt.show()
```

Program 4: Scatter Chart

PYTHON

```
import matplotlib.pyplot as plt
plt.scatter([2,9,8,5,6], [5,10,3,7,18], color="red", marker="*")
plt.xlabel("X"); plt.ylabel("Y"); plt.title("Scatter Chart"); plt.show()
```

Program 5: Read CSV & Display 10 Rows

PYTHON

```
import pandas as pd
df = pd.read_csv("data.csv")
print(df.head(10))
```

Program 6: Read CSV & Display Information

PYTHON

```
import pandas as pd
df = pd.read_csv("data.csv")
print("Shape:", df.shape); print(df.dtypes); df.info(); print(df.describe())
```

Program 7: Read & Display Image

PYTHON

```
import cv2
img = cv2.imread("sample.jpg")
cv2.imshow("My Image", img); cv2.waitKey(0); cv2.destroyAllWindows()
```

Program 8: Read Image & Identify Shape

PYTHON

```
import cv2
img = cv2.imread("sample.jpg")
print("Shape:", img.shape)   # (height, width, channels)
print("Height:", img.shape[0], "Width:", img.shape[1], "Channels:", img.shape[2])
```

Program 9: NumPy Array Operations

PYTHON

```python
import numpy as np
a = np.array([10,20,30]); b = np.array([1,2,3])
print("Add:", a+b, "Sub:", a-b, "Mul:", a*b)  # [11,22,33] [9,18,27] [10,40,90]
```

Program 10: Create DataFrame & Statistics

PYTHON

```python
import pandas as pd
data = {"Name": ["Aman","Priya","Rahul"], "Marks": [85,92,67]}
df = pd.DataFrame(data)
print(df); print(df.describe()); df.info()
```

Program 11: Bar Chart

PYTHON

```python
import matplotlib.pyplot as plt
plt.bar(["Math","Sci","Eng","AI"], [85,90,78,95], color=["red","blue","green","purple"])
plt.xlabel("Subject"); plt.ylabel("Marks"); plt.title("Marks"); plt.show()
```

Program 12: Pie Chart

PYTHON

```python
import matplotlib.pyplot as plt
plt.pie([35,25,20,20], labels=["Math","Sci","AI","Eng"], autopct="%1.1f%%")
plt.title("Subject Distribution"); plt.show()
```

Program 13: Convert to Grayscale

PYTHON

```python
import cv2
img = cv2.imread("sample.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
print("Original:", img.shape, "Gray:", gray.shape)
cv2.imshow("Gray", gray); cv2.waitKey(0); cv2.destroyAllWindows()
```

Program 14: Resize Image

PYTHON

```
import cv2
img = cv2.imread("sample.jpg")
resized = cv2.resize(img, (300, 200))  # (width, height)
print("Resized:", resized.shape)  # (200, 300, 3)
cv2.imshow("Resized", resized); cv2.waitKey(0); cv2.destroyAllWindows()
```

Program 15: Draw Shapes

PYTHON

```
import cv2; import numpy as np
img = np.zeros((500,500,3), dtype=np.uint8); img[:] = (255,255,255)
cv2.line(img,(50,50),(450,50),(255,0,0),3)  # Blue line
cv2.rectangle(img,(50,100),(250,250),(0,255,0),3)  # Green rect
cv2.circle(img,(375,175),75,(0,0,255),-1)  # Red filled circle
cv2.putText(img,"AI Class 10",(100,350),cv2.FONT_HERSHEY_SIMPLEX,1.5,(0,0,0),3)
cv2.imshow("Shapes",img); cv2.waitKey(0); cv2.destroyAllWindows()
```

Program 16: Histogram

PYTHON

```
import matplotlib.pyplot as plt
plt.hist([45,67,78,85,92,55,73,88,62,95], bins=5, color="skyblue", edgecolor="black")
plt.xlabel("Marks"); plt.ylabel("Students"); plt.title("Distribution"); plt.show()
```

Program 17: Multiple Line Charts

PYTHON

```
import matplotlib.pyplot as plt
sub = ["Math","Sci","Eng","AI","Hindi"]
plt.plot(sub,[85,78,90,95,72],marker="o",label="Aman")
plt.plot(sub,[92,88,82,89,95],marker="s",linestyle="--",label="Priya")
plt.legend(); plt.title("Comparison"); plt.show()
```

Program 18: Filter DataFrame

PYTHON

```
import pandas as pd
data = {"Name":["Aman","Priya","Rahul","Sneha"], "Marks":[85,92,45,78]}
df = pd.DataFrame(data)
toppers = df[df["Marks"] > 75]
print("Toppers:"); print(toppers)
```

C.3 Sample Viva Questions with Answers

V1 What is AI?

AI is the simulation of human intelligence in machines that can learn, reason, and make decisions.

V2 Name the three AI domains.

Data Science, Computer Vision (CV), Natural Language Processing (NLP).

V3 What are the 5 stages of AI Project Cycle?

Problem Scoping → Data Acquisition → Data Exploration → Modelling → Evaluation.

V4 What is a variable?

A named container that stores a value in memory. Example: `x = 10`.

V5 Difference between list and tuple?

List is mutable (changeable), uses []. Tuple is immutable (fixed), uses (). Lists slower but flexible; tuples faster but fixed.

V6 What does input() return?

Always returns a string. Must convert with int() or float() for math.

V7 What is a pixel?

Smallest unit of a digital image. Grayscale: 1 value (0-255). RGB: 3 values (R,G,B each 0-255).

V8 Why does OpenCV use BGR?

Historical convention from early camera hardware. Convert to RGB with cv2.cvtColor(img, cv2.COLOR_BGR2RGB).

V9 What is mean vs median?

Mean = average (sum/count). Median = middle value when sorted. Median is better with outliers.

V10 What is tokenization?

Breaking text into smaller units (words or sentences). "AI is fun" → ["AI", "is", "fun"].

V11 What is a confusion matrix?

A 2×2 table showing TP, TN, FP, FN to evaluate AI model performance.

V12 What is OpenCV?

Open Source Computer Vision library for image processing. Imported as cv2.

V13 What does df.describe() show?

Statistics: count, mean, std, min, 25%, 50%, 75%, max for numeric columns.

V14 What is Bag of Words?

Text representation that counts word frequency, ignoring grammar and order.

V15 What is accuracy in AI evaluation?

Accuracy = (TP+TN)/(TP+TN+FP+FN). Percentage of correct predictions out of total.

Before Showing Output: "RICE" Check

Run the code • **I**nspect output • **C**heck for errors • **E**xplain each line

D

Project Work, Field Visit & Student Portfolio

AI project report, field visit documentation, portfolio guidelines, and viva preparation.

PART D   Project/Portfolio: 10 marks   Viva: 5 marks

D.1 Marks Distribution

Component	Details	Marks
Project / Field Visit / Portfolio	Problem + SDG + AI Project Cycle + Documentation	10
Viva Voce	Understanding of project + AI concepts	5
Total		**15**

D.2 Sample Project: Student Marks Prediction

Complete AI Project Cycle Walkthrough

SDG: SDG 4 — Quality Education

1. Problem Scoping (4Ws):
WHO: Teachers, students. WHAT: Can't identify at-risk students before exams. WHERE: School classrooms. WHY: Early intervention improves pass rates.

2. Data Acquisition:
Data: Study hours, attendance %, previous marks, homework completion. Source: School database + student survey. Format: CSV, 30-50 records.

3. Data Exploration:
Scatter plot shows positive correlation between study hours and marks. Students with >85% attendance rarely fail. Used Pandas describe() and Matplotlib charts.

4. Modelling:
Supervised Learning with Linear Regression. Input: study hours, attendance, previous marks. Output: predicted final marks.

5. Evaluation:
Compare predicted vs actual marks. If average error <10 marks, model is useful. Teacher feedback confirms predictions match their observations.

Sample Project 2: CNN Model on Smoke and Fire Detection

Sustainable Development Goals:

- **SDG 11: Sustainable Cities and Communities** — Protecting residential buildings, schools, and smart cities from fire disasters.
- **SDG 15: Life on Land** — Detecting forest fires early to protect wildlife, trees, and biodiversity.

1. Problem Scoping (4Ws):

- *Who is affected?* Residents, fire departments, forest rangers, and wildlife.
- *What is the problem?* Delay in fire detection leads to uncontrollable fires, causing destruction of property and habitats.
- *Where does it occur?* In forests, commercial buildings, smart homes, and industrial plants.

- *Why solve it?* Real-time automated detection allows immediate alerts, enabling containment before the fire spreads out of control.

2. Data Acquisition:

- **Data Types:** Image and video frames of smoke, fire, and normal environments (no fire).
- **Source:** Custom camera feeds, drone footage, or public datasets (e.g., FIRED Dataset, Kaggle Fire Detection Dataset).
- **Dataset size:** 2,000 labeled images (1,000 "Fire/Smoke" and 1,000 "Normal").

3. Data Exploration:

- Check aspect ratios and image resolutions.
- Understand color distribution: Fire features high intensity values in Red and Yellow channels; smoke contains greyish, hazy textures.
- Normalize images to a standard size (128×128 pixels) and scale pixel values between 0 and 1.

4. Modelling:

- **Type of AI:** Computer Vision (Deep Learning).
- **Algorithm:** Convolutional Neural Network (CNN).
- **Architecture:**
  - *Input Layer:* Receives $128 \times 128 \times 3$ RGB images.
  - *Conv2D + MaxPooling Layers:* Extract geometric edges, color boundaries, and texture features of flames and smoke.
  - *Flatten Layer:* flattens the multi-dimensional feature map into a 1D vector.
  - *Dense Layers + Softmax/Sigmoid Output:* Classify the image into **"Fire Detected"** or **"Safe/Normal"**.

5. Evaluation:

- Compare model predictions against test images.
- **Key Metric: Recall** is critical because a False Negative (failing to detect a real fire) can be catastrophic. We aim for a high Recall (e.g., >98%) to ensure no actual fire is missed.

D.3 Project Report Format

1. Title Page (School name, project title, student name, SDG number)
2. Certificate (signed by teacher and principal)
3. Acknowledgment
4. Table of Contents
5. Introduction / Problem Statement
6. SDG Mapping
7. AI Project Cycle — all 5 stages documented
8. Code / Implementation (with screenshots)
9. Output / Results (with charts)
10. Conclusion
11. Bibliography

Tips: 15-20 pages. A4 paper. Use headings. Include charts. Write in own words. Add page numbers. Bind neatly.

D.4 Field Visit & Portfolio

Field Work / Visit: Students must participate in at least one activity to get acquainted with real-life AI usage and document it (Date, Location, Objective, AI Technologies observed, SDG connection, and Learning outcomes):

- **AI for Youth Bootcamp:** Participatory bootcamps conducted by CBSE/Intel or other partners to learn AI development.
- **AI Fests / Exhibitions:** School, regional, or national fests where students showcase AI solutions or learn from peers.
- **AI Training Sessions:** Educational workshops or seminars led by industry professionals.
- **Virtual Company Tours:** Virtual visits to tech hubs, smart farms, hospitals, or AI laboratories using AI tools.

Student Portfolio: A cumulative record continued from Class IX that showcases a student's learning journey and technical skills. **Note: A portfolio must contain a minimum of 5 activities.**

- **Activity Records:** Maintaining a detailed, chronological record of all AI activities and practical sheets completed in class.
- **Hackathons:** Coding challenges or prototype-building sessions.
- **Competitions:** Participation in board-level (CBSE) or Interschool AI competitions.
- **Certificates & Courses:** Badges, certifications, or class presentations.

CBSE Assessment Note

NOTE: Optional curriculum components (such as K-Nearest Neighbour model, TF-IDF calculation steps, NLTK python coding, CNN operator, and CNN testing details) **shall not be assessed** in formal school theory examinations. They are provided solely for extra knowledge, practical projects, and competitive preparation.

D.5 Project Viva Questions

V1 What is your project title?

State clearly. Example: "Student Marks Prediction System."

V2 Which SDG does your project relate to?

State SDG number and name. Example: "SDG 4 — Quality Education."

V3 What problem does your project solve?

Explain the real-world problem. Example: "Teachers can't identify struggling students early enough."

V4 What are the stages of AI Project Cycle?

Problem Scoping → Data Acquisition → Data Exploration → Modelling → Evaluation. "Please Don't Delay My Exam."

V5 What type of ML model would you use?

Supervised Learning — because we have labelled data (known marks) and want to predict a number (marks).

D.6 Complete Marks Summary

Here is the official unit-wise marks distribution and weightage guidelines for the CBSE Class 10 Artificial Intelligence board exam (Subject Code 417):

Part	Syllabus Unit & Sub-Unit	Assessment Domain	Theory Marks	Practical Marks	Total Marks
Part A: Employability Skills	Unit 1: Communication Skills-II	Theory	2	-	**10**
	Unit 2: Self-Management Skills-II	Theory	2	-	
	Unit 3: ICT Skills-II	Theory	2	-	
	Unit 4: Entrepreneurial Skills-II	Theory	2	-	
	Unit 5: Green Skills-II	Theory	2	-	
Part B: Subject Specific Skills	Unit 1: Introduction to AI	Theory	7	-	**40**
	Unit 2: AI Project Cycle	Theory	9	-	
	Unit 3: Advance Python	Practical Only	-	-	
	Unit 4: Data Science	Theory & Practical	4	-	
	Unit 5: Computer Vision	Theory & Practical	4	-	

	Unit 6: Natural Language Processing	Theory & Practical	8	-	
	Unit 7: Evaluation	Theory	8	-	
Part C: Practical Work	Practical File (Minimum 15 Programs)	Practical	-	15	**35**
	Practical Exam (Python: 5, DS: 5, CV: 5)	Practical	-	15	
	Viva Voce	Practical	-	5	
Part D: Project Work & Portfolio	Project Work / Field Visit / Student Portfolio	Practical	-	10	**15**
	Viva Voce	Practical	-	5	
GRAND TOTAL		**Theory & Practical**	**50 Marks**	**50 Marks**	**100 Marks**

D.7 Quick Revision — All Memory Tricks

AI Project Cycle: "Please Don't Delay My Exam"
Problem Scoping → Data Acquisition → Data Exploration → Modelling → Evaluation

AI Domains: "Data Needs Cameras"
Data Science → NLP → Computer Vision

ML Types: "SUR"
Supervised (Teacher) → Unsupervised (Self) → Reinforcement (Reward)

Evaluation: "All People Read Fast"
Accuracy → Precision → Recall → F1 Score

NLP Steps: "Teachers Should Study Lessons Carefully"
Tokenization → Stop words → Stemming → Lemmatization → Case normalization

OpenCV: "Read, Show, Wait, Destroy"
imread() → imshow() → waitKey(0) → destroyAllWindows()